FOR EDITH · OBSTETRICS & GYNAECOLOGY JUNIOR CLERKSHIP

Infertility

Tutorial | Week 5 · Bleeding, cancer & fertility | Gynaecology | Prof Mirembe

Infertility is failure to achieve a clinical pregnancy after 12 months of regular unprotected intercourse; it is a problem of the couple, evaluated systematically by asking four questions — is she ovulating, are the tubes open, is the uterine cavity normal, and is the semen normal — and in Uganda it is dominated by secondary, infection-related tubal damage and carries a heavy social cost.

Must know before the ward round

1. Definition (WHO) — failure to achieve a clinical pregnancy after ≥12 months of regular unprotected intercourse; investigate after 6 months if the woman is 35 or older or there is an obvious cause.
2. About 80% of couples conceive within 1 year and about 90% within 2 years; female fertility falls from the early 30s and steeply after 37.
3. In Uganda secondary infertility predominates and tubal blockage (after STIs/PID, post-abortal and post-partum sepsis) is the commonest finding on HSG.
4. Four pillars of evaluation — ovulation (mid-luteal progesterone about 7 days before the next period; >30 nmol/L confirms ovulation), tubes (HSG in days 6–10 of the cycle), uterus (ultrasound/HSG/hysteroscopy), semen (WHO 2021 lower reference limits — volume 1.4 mL, 16 million/mL, total 39 million, total motility 42%, progressive 30%, normal forms 4%, vitality 54%).
5. PCOS (Rotterdam, 2 of 3) — oligo/anovulation, clinical or biochemical hyperandrogenism, polycystic ovaries on ultrasound (≥20 follicles per ovary) or raised AMH in adults — after excluding other causes.
6. Letrozole 2.5 mg daily on days 3–7 (up to 7.5 mg) is first-line ovulation induction in PCOS; clomiphene 50 mg daily on days 2–6 (up to 150 mg), with ultrasound monitoring and for no more than 6 months, is the alternative.
7. Never start ovulation induction until you know the tubes are patent and the semen is adequate.
8. Hydrosalpinx halves IVF success — salpingectomy or proximal tubal occlusion before IVF.
9. Semen analysis comes first and is the cheapest test — 2–7 days of abstinence, to the lab within 1 hour at body temperature; repeat an abnormal result after about 3 months (spermatogenesis takes about 74 days).
10. Time the mid-luteal progesterone 7 days before the expected period (day 28 of a 35-day cycle); a low "day-21" value in a long cycle is usually mistimed, so repeat before calling it anovulation.
11. Oligo/amenorrhoea needs hormones, not a progesterone — FSH, LH and oestradiol on days 2–5, prolactin, TSH, testosterone and a pregnancy test; FSH above 25 IU/L (repeated) under 40 means POI, low FSH/LH/E2 means WHO group I.
12. On HSG, free bilateral spill means patent tubes; club-shaped tubes without spill are hydrosalpinges (IVF after salpingectomy); cornual block may be spasm (confirm by laparoscopy and dye); beaded pipe-stem tubes suggest genital TB.

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1. Definition

Infertility is a disease of the couple, not of the woman. In Uganda (part of Africa's "infertility belt") the commonest story is a woman who had one or two children (or one pregnancy that ended in an abortion or a septic delivery) and now cannot conceive: **secondary infertility from tubal damage** caused by untreated gonorrhoea and chlamydia, post-abortal sepsis and post-partum infection. Men are rarely examined, even though male factors contribute to about half of all cases.

Term	Definition
Infertility (WHO/ICMART)	Failure to achieve a **clinical pregnancy after ≥12 months of regular unprotected sexual intercourse**
Primary infertility	The woman (or couple) has **never** conceived
Secondary infertility	Previous pregnancy (whatever the outcome: live birth, miscarriage, abortion, ectopic) but now unable to conceive
Subfertility	Often used interchangeably; emphasises reduced, not absent, fertility
Fecundability	Probability of conception per menstrual cycle: about **20%** in young couples
Sterility	Absolute inability to conceive (e.g. bilateral salpingectomy, azoospermia with no sperm production)
Unexplained infertility	Normal ovulation, patent tubes, normal cavity and normal semen analysis
Recurrent pregnancy loss	A separate problem: the couple conceives but loses the pregnancies

When to start investigating: after **12 months** of trying; **after 6 months** (or immediately) if the woman is **≥35 years**, or there is an obvious risk factor: amenorrhoea/oligomenorrhoea, previous PID/ectopic/pelvic surgery, known endometriosis, previous chemotherapy/radiotherapy, known male problem (undescended testes, mumps orchitis, erectile dysfunction), or HIV discordance planning.

Evaluate every couple with **four questions**:

1. **Is she ovulating?** (history, mid-luteal progesterone, hormones)
2. **Are the tubes open?** (HSG, laparoscopy and dye)
3. **Is the uterine cavity normal?** (ultrasound, HSG, hysteroscopy)
4. **Is the semen normal?** (semen analysis: the cheapest, most neglected test)

Then treat the cause: induce ovulation, open or bypass the tubes, fix the cavity, or treat or bypass the male factor, and **counsel**, because the emotional and social harm (blame, divorce, violence, stigma) is often worse than the medical problem.

> **RED FLAG**
> **The one thing never to forget:** do **not** give clomiphene or letrozole "to try" before a **semen analysis** and **tubal assessment**. Ovulation induction with blocked tubes or azoospermia wastes months of her fertile life, risks ectopic pregnancy and multiple pregnancy, and delays the correct treatment.

2. Epidemiology

- WHO (2023) estimates that about **1 in 6 people** worldwide experience infertility in their lifetime.
- About **80%** of couples conceive within 1 year and about **90%** within 2 years; female fertility falls from the early 30s and steeply after 37.
- In Uganda, infertility is common and **secondary infertility is more common than primary**. A study of 289 consecutive HSGs in Kampala found an **abnormality in 83%**, the commonest being **tubal blockage**, attributed to chronic pelvic infection Verify *citation*.
- WHO studies attributed the large majority of female infertility in sub-Saharan Africa to prior infection.
- **Services**: basic assessment (history, examination, semen analysis, HSG, ultrasound, hormones) is available at **regional and national referral hospitals** and many private clinics; laparoscopy and hysteroscopy are limited to a few centres. **IVF** has been available in **private centres in Kampala since the first IVF birth in 2004**; **Mulago Specialised Women and Neonatal Hospital** started the first **public IVF service in 2025** Verify *dates*. Costs are out of reach for most families, and there is no insurance cover.
- **Social reality**: a childless woman is often blamed, may be abandoned, face polygamy, divorce, violence or disinheritance, and may spend years and savings on traditional healers and repeated "tuboplasties". Men often refuse semen analysis.

> UGANDA CONTEXT
> The most effective "fertility treatment" in Uganda is **prevention**: early syndromic treatment of STIs **for both partners**, **safe post-abortion care** (MVA and antibiotics), **clean deliveries and caesarean sections with prophylactic antibiotics**, and early treatment of PID. Every time you manage a woman with PID or post-abortal sepsis properly, you are preventing infertility.

3. Aetiology and Risk Factors

Causes (approximate contribution in a clinic population)

Cause	Approximate share	Uganda note
Male factor (alone or combined)	30–50%	Under-investigated because of stigma
Tubal / peritoneal	20–35% globally	**Much higher** in sub-Saharan Africa: STI/PID, post-abortal and post-partum sepsis, genital TB
Ovulatory	About 25%	PCOS commonest; hyperprolactinaemia, thyroid, low weight, POI
Uterine / cervical	About 10%	Fibroids (very common in Ugandan women), Asherman's after D&C/sepsis
Endometriosis	5–10%	Probably under-diagnosed (laparoscopy rarely available)
Unexplained	About 25%	

Drivers of tubal infertility in Uganda: untreated/under-treated gonorrhoea and chlamydia (often asymptomatic), **unsafe abortion and post-abortal sepsis**, **post-partum and post-caesarean infection**, **genital tuberculosis**, and schistosomiasis in some regions.

Male factor risk factors: STIs (epididymo-orchitis), mumps, varicocele, alcohol, smoking, heat, anabolic steroids, HIV and its treatment, and diabetes.

Male causes by level

- **Pre-testicular**: hypogonadotropic hypogonadism; exogenous testosterone/anabolic steroids (which **suppress** FSH/LH and switch off spermatogenesis).
- **Testicular**: mumps orchitis, cryptorchidism, varicocele, Klinefelter syndrome, chemotherapy, heat.
- **Post-testicular**: obstruction after epididymitis/gonorrhoea, congenital absence of the vas in CF carriers, vasectomy, retrograde ejaculation in diabetes, erectile dysfunction.

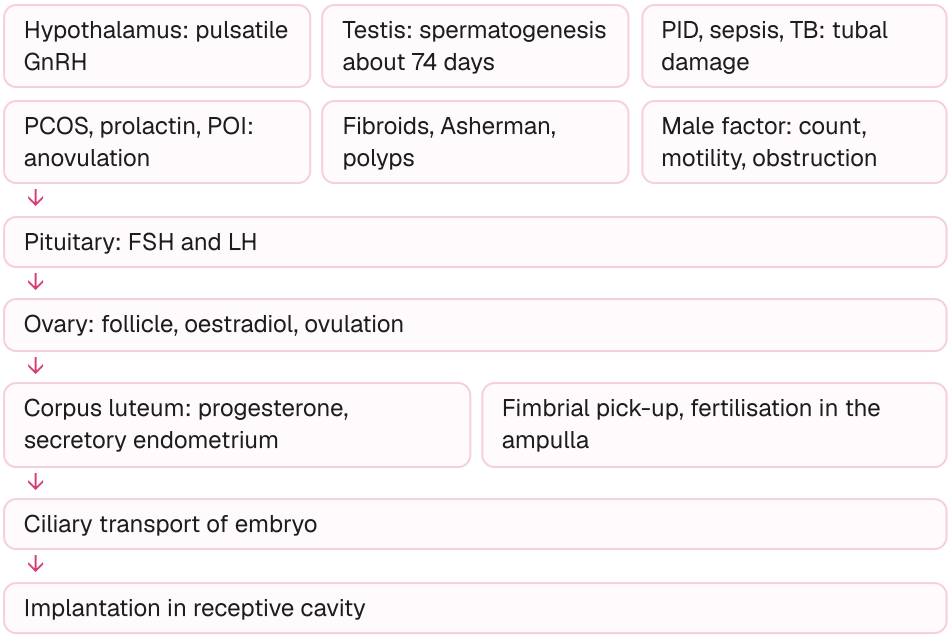
Other female risk factors: age above 35; BMI below 19 or above 30; smoking; previous pelvic surgery (perforated appendicectomy, ectopic, myomectomy, CS, ovarian cystectomy); drugs raising prolactin (antipsychotics, metoclopramide); recent DMPA (return of fertility can take up to about 10 months after the last injection); coital problems.

4. Pathophysiology

Normal conception

GnRH from the hypothalamus is released in pulses; the pituitary releases FSH and LH; the ovary makes oestradiol and progesterone, which act on the endometrium and feed back on the hypothalamus and pituitary (negative and positive feedback).

1. **Follicular phase** (day 1 to about day 14): FSH recruits follicles; one dominant follicle grows to about **18–25 mm** and produces **oestradiol**, which thickens the endometrium and makes cervical mucus thin and stretchy (sperm-friendly).
2. **Ovulation**: rising oestradiol triggers the **LH surge** (positive feedback); ovulation follows about **36 hours** after the start of the surge.
3. **Luteal phase** (fixed about **14 days**): the corpus luteum secretes **progesterone**, which prepares a secretory endometrium; progesterone **peaks about 7 days after ovulation**. That is why we measure it **7 days before the expected period** ("day 21" of a 28-day cycle; **day 28 of a 35-day cycle**).
4. **Fertile window**: sperm survive **3–5 days**, the oocyte **12–24 hours**, so the fertile window is the **5 days before and the day of ovulation**. Hence advice: intercourse **every 2–3 days** throughout the cycle (timed intercourse causes stress and is not needed).
5. **Sperm transport and fertilisation**: sperm pass cervical mucus, undergo **capacitation**, and meet the oocyte in the **ampulla** of the tube; the fimbriae must be free to pick up the oocyte and the **ciliated tubal epithelium** must transport the embryo to the uterus (about 5 days).
6. **Implantation** at about day 6–7 after fertilisation in a receptive endometrium.
7. **Spermatogenesis** takes about **74 days** (about 3 months including transport), so a semen analysis reflects health 3 months ago, and lifestyle changes take 3 months to show.



Where things go wrong

- **Tubal damage (PID)**: *C. trachomatis* and *N. gonorrhoeae* ascend, causing salpingitis; inflammation destroys **cilia**, causes intraluminal adhesions and **fimbrial agglutination**, leading to **distal occlusion** and **hydrosalpinx** (fluid-filled tube). Partial damage lets sperm through but traps the embryo → **ectopic pregnancy**. Peri-tubal adhesions (after sepsis or surgery) prevent oocyte pick-up. Each episode of PID roughly doubles the risk of tubal infertility.
- **Hydrosalpinx fluid** leaks back into the uterine cavity: it is **embryotoxic** and mechanically washes out embryos, hence it **halves IVF success** unless removed/clipped.
- **PCOS**: insulin resistance and high insulin → ovarian theca cells over-produce **androgens**; high LH pulse frequency; many small follicles arrest at 2–9 mm, no dominant follicle → **anovulation**. Androgens cause **hirsutism and acne**; unopposed oestrogen thickens the endometrium (**endometrial hyperplasia** risk).
- **Hyperprolactinaemia**: prolactin inhibits pulsatile **GnRH** → low FSH/LH → anovulation, amenorrhoea, galactorrhoea.
- **Asherman's syndrome**: vigorous curettage of a **recently pregnant (soft) uterus** or post-abortal/post-partum infection removes the basal endometrium → intrauterine adhesions → scanty periods/amenorrhoea, infertility, miscarriage.
- **Fibroids**: submucous (and large intramural) fibroids distort the cavity and impair implantation.
- **Endometriosis**: inflammation, adhesions, distorted anatomy, reduced oocyte quality.
- **Male**: pre-testicular, testicular and post-testicular failure (section 3).

Feature links

- Anovulation → no corpus luteum → irregular or absent periods, low mid-luteal progesterone.
- PCOS hyperinsulinaemia → ovarian androgens → hirsutism, acne; insulin resistance → acanthosis nigricans, obesity.
- Hyperprolactinaemia → GnRH suppression → amenorrhoea; prolactin → galactorrhoea; macroadenoma → headache, bitemporal field loss.
- Ovarian failure (POI) → low oestradiol, loss of feedback → high FSH, hot flushes, amenorrhoea under 40.
- Hypothalamic failure (low weight, Sheehan's) → low FSH, LH and oestradiol; Sheehan's after severe PPH → failure of lactation, amenorrhoea.
- Salpingitis → cilia loss, fimbrial agglutination → hydrosalpinx (club-shaped tubes, no spill on HSG; tubular adnexal fluid on ultrasound) and ectopic pregnancy.

- Pelvic adhesions (PID, endometriosis) → fixed retroverted uterus, adnexal thickening; endometriosis → progressive dysmenorrhoea, deep dyspareunia, nodules in the pouch of Douglas.
- Genital TB → tubal and endometrial destruction → beaded pipe-stem tubes, scanty periods.
- Curettage or sepsis → basal endometrium lost → Asherman's (scanty periods, irregular cavity on HSG).
- Submucous fibroid → cavity distortion → heavy periods, filling defect on HSG.
- Testicular failure → small soft testes, high FSH, oligo- or azoospermia; obstruction → normal testes, normal FSH, azoospermia; absent vasa → low-volume azoospermia (CBAVD).
- Exogenous testosterone → negative feedback → low FSH/LH → spermatogenesis switched off.

5. Classification / Staging

Primary versus secondary infertility (section 1).

WHO classification of anovulation (asked in vivas)

WHO group	Problem	Hormones	Examples	Treatment principle
I: hypogonadotropic hypogonadism	Hypothalamus/pituitary fails	**Low FSH, LH, oestradiol**	Low weight, excessive exercise, stress, Sheehan's syndrome (after PPH), Kallmann syndrome, pituitary tumour	Weight gain; **gonadotrophins with LH activity** or pulsatile GnRH (clomiphene/letrozole do not work: nothing to "unblock")
II: normogonadotropic, normo-oestrogenic	Dysfunction of the axis	**Normal FSH, normal E2** (LH often high)	**PCOS** (80–90% of anovulation)	Weight loss, **letrozole**, clomiphene ± metformin, gonadotrophins, ovarian drilling
III: hypergonadotropic hypogonadism	**Ovary fails**	**High FSH**, low E2	**Premature ovarian insufficiency** (<40 y), post-chemotherapy, Turner syndrome	Donor oocytes (IVF); HRT for health
(Hyperprolactinaemia, often listed separately)	Prolactin suppresses GnRH	High prolactin	Prolactinoma, drugs (antipsychotics, metoclopramide), hypothyroidism	Dopamine agonist (cabergoline/bromocriptine)

PCOS (Rotterdam, 2 of 3), after excluding other causes: oligo/anovulation; clinical or biochemical hyperandrogenism; polycystic ovarian morphology on ultrasound (**≥20 follicles per ovary** or ovarian volume **≥10 mL**) or raised AMH in adults.

Semen analysis terminology: **oligozoospermia** (low count), **asthenozoospermia** (low motility), **teratozoospermia** (abnormal forms), **azoospermia** (no sperm: obstructive or non-obstructive), **aspermia** (no ejaculate).

6. Clinical Features

See the couple together, then each alone (confidential questions: previous pregnancies with other partners, STIs, HIV, violence).

History

The woman

Ask	Why: the answer that changes the plan
Age	The single strongest predictor of success; ≥35 → investigate at 6 months and move faster; at 38–40 refer early for ovarian reserve and ART
Duration of trying, frequency of intercourse, partner living together?	12 months or more → investigate. Many "infertile" couples live apart (migrant work) or have infrequent sex
Primary or secondary? Outcome of each pregnancy	Secondary after abortion, D&C, septic delivery or CS → **tubal or Asherman's**: HSG early
Previous **abortion, D&C, MVA, post-partum fever, PPH, retained placenta**	Sepsis → tubal damage; curettage → Asherman's; severe PPH → Sheehan's (no lactation, amenorrhoea)
STIs, PID, lower abdominal pain, vaginal discharge, treatment of partner	Tubal damage → HSG ± laparoscopy
Menstrual history: cycle length/regularity, amenorrhoea, scanty periods, heavy periods, dysmenorrhoea	**21–35 days and regular** → probably ovulating: confirm with mid-luteal progesterone **timed to her cycle**. **Oligo/amenorrhoea** → skip the progesterone and do **hormones**. Scanty after D&C → Asherman's; heavy → fibroids/polyps; progressive dysmenorrhoea + deep dyspareunia → endometriosis (laparoscopy)
Hirsutism, acne, weight gain	PCOS → testosterone, ultrasound, glucose
Galactorrhoea, headache, visual disturbance	Prolactinoma
Hot flushes, amenorrhoea <40 years	Premature ovarian insufficiency (FSH)
Weight change, exercise, eating	Hypothalamic amenorrhoea (WHO I)
Thyroid symptoms	Hypo-/hyperthyroidism disturbs ovulation
Previous surgery: appendicectomy (perforated), ectopic, myomectomy, CS, ovarian cystectomy	Adhesions, tubal damage, reduced ovarian reserve
TB (history, contacts), **schistosomiasis**	Genital TB: tubal and endometrial destruction → endometrial biopsy
Contraception history (DMPA especially)	DMPA can delay return of fertility for up to about 10 months after the last injection: may simply need time
Drugs (antipsychotics, metoclopramide → prolactin), herbal medicines, smoking, alcohol	Reversible factors
HIV status (both partners)	Safer conception planning; ART interactions
Coital problems: dyspareunia, vaginismus, erectile/ejaculatory difficulties	Some couples never have complete intercourse; needs different care
Psychological and social impact; violence; what has been tried (herbs, healers, previous "tuboplasty"); money available	Counselling needs; safety; realistic plan; honest counselling about IVF costs

The man

Ask	Why it matters
Has he fathered a child before?	Previous fertility does not exclude a new problem
Mumps orchitis after puberty, **undescended testes**, testicular trauma/torsion	Testicular failure
Urethral discharge, epididymo-orchitis, STI treatment	Obstructive azoospermia (epididymal/vasal blockage)
Hernia repair, scrotal surgery, vasectomy	Vas injury/obstruction
Erectile or ejaculatory dysfunction, low-volume or no ejaculate	Coital failure, retrograde ejaculation (diabetes)
Anabolic steroids / testosterone, chemotherapy, sulfasalazine, recreational drugs	Suppressed or damaged spermatogenesis (often reversible)
Heat (drivers, welders, bakers), tight clothing, laptops on lap, smoking, heavy alcohol	Reversible lifestyle factors
Diabetes, HIV, chronic disease, cystic fibrosis symptoms (chest disease)	Systemic causes; CBAVD

Examination

Follow the order of Hutchison's Clinical Methods (24th edition). Examine **both** partners; do not assume.

Introduction, consent and preparation

- Introduce yourself to both partners, explain confidentiality, explain each examination and gain consent; chaperone for breast, pelvic and genital examination; examine each partner privately.
- **Hand hygiene**.
- **Position and exposure**: woman supine for the abdomen, then lithotomy or dorsal position for speculum and bimanual; man standing for the scrotum (varicocele), then supine.

The woman

- **General inspection and vital signs**: **BMI** (<19 or >30), waist circumference, **BP**; secondary sexual characteristics (breast, pubic hair: Tanner stage).
- **Hands, face and neck**: **hirsutism** (upper lip, chin, chest, linea alba), acne, **acanthosis nigricans** (neck, axillae: insulin resistance); **thyroid**; **visual fields** if prolactin is likely high.
- **Breasts**: **galactorrhoea** (express gently).
- **Abdomen**: scars (CS, laparotomy, appendicectomy), masses (fibroid uterus, ovarian mass), ascites.
- **Pelvis**: vulva and vagina (vaginismus, septum, clitoromegaly); **speculum** (cervix, discharge, **VIA** if due); **bimanual**: uterine size and contour (**fibroids**: bulky irregular uterus), **mobility** (a **fixed retroverted uterus** suggests adhesions from PID or endometriosis), **adnexal thickening or tenderness**, adnexal mass, tender nodules in the pouch of Douglas (endometriosis).
- **Document key negatives**: "no hirsutism, no galactorrhoea, uterus normal size, mobile, adnexa free".

The man

- **General**: virilisation, **gynaecomastia** (Klinefelter, steroids), BMI.
- **Testes**: volume with an orchidometer (normal about **15–25 mL**; small soft testes = testicular failure), consistency, masses.
- **Vas deferens**: palpable **on both sides**? (absent = congenital bilateral absence of the vas, CF gene).
- **Epididymis**: thickening or cysts (previous epididymitis, obstruction).
- **Varicocele**: examine **standing** with Valsalva ("bag of worms").
- **Penis**: hypospadias, discharge; hernia scars.

Completing the examination

Thank both, hand hygiene, summarise to the couple together, and outline the four-part investigation plan.

Red flags

> **RED FLAG**
>
> - **Positive pregnancy test with pain or bleeding** in a woman "trying for years" or treated for infertility → **ectopic until proven otherwise** (damaged tubes): vitals, ultrasound, senior review today.
> - **Abdominal distension, pain, breathlessness, vomiting, oliguria after ovulation induction/IVF** → ovarian hyperstimulation syndrome (OHSS).
> - **Galactorrhoea with headache or visual field loss** → pituitary macroadenoma: MRI and endocrine referral.
> - **Amenorrhoea with hot flushes under 40** → premature ovarian insufficiency: confirm and counsel early (time-critical).
> - **Rapid-onset hirsutism/virilisation** (clitoromegaly, deep voice) → androgen-secreting tumour, not PCOS.
> - **Pelvic mass**, weight loss, ascites → ovarian tumour.
> - **Testicular mass** → testicular cancer (infertile men have a higher risk).
> - **Azoospermia** → full male work-up (hormones, karyotype): never just "give the wife clomiphene".
> - **Depression, suicidal ideation, intimate partner violence**: ask directly.

Thinking like a clinician

- **Secondary infertility + history of abortion/post-partum sepsis/PID = tubes until proven otherwise** → HSG early.
- **Scanty periods after a D&C** = Asherman's until proven otherwise → HSG/hysteroscopy.
- **Oligomenorrhoea + hirsutism + obesity** = PCOS, but exclude thyroid disease, hyperprolactinaemia and (if very high androgens) tumours or non-classic CAH.
- **Regular cycles (every 21–35 days) almost always mean ovulation**: still confirm with mid-luteal progesterone, but look hard at tubes and semen.
- **Mid-luteal progesterone must be timed to the cycle**: 7 days **before** the expected period, not "day 21" in a woman with a 35-day cycle; a "low day-21 progesterone" in a long cycle is often just mistimed.
- **The man first?** A semen analysis is cheap, non-invasive and should be done **before** the woman has an HSG or laparoscopy. Engage men respectfully: invite them to the clinic, explain that half of infertility involves men.
- **Age changes everything**: a 38-year-old should not spend two years on clomiphene; refer early for assessment of ovarian reserve and ART.
- **Common student mistakes**: presenting infertility with no male history; calling a 35-day cycle "irregular"; ordering "day-21 progesterone" in a woman with amenorrhoea; forgetting a pregnancy test; recommending ovulation induction without tubal and semen assessment; forgetting HIV/serodiscordance and preconception folic acid; forgetting the emotional and social history.

7. Differential Diagnoses

Differential diagnosis **of the cause**:

Differential	Distinguishing clinical feature	Distinguishing investigation
Tubal factor (post-infection)	Secondary infertility, previous PID/abortion/puerperal sepsis, regular cycles, adnexal tenderness/thickening	**HSG** (blocked tubes, hydrosalpinx); laparoscopy and dye

Differential	Distinguishing clinical feature	Distinguishing investigation
Genital tuberculosis	TB history/contact, amenorrhoea/scanty periods, "beaded" rigid tubes, calcification on X-ray	HSG (beaded tubes, "pipe-stem"); **endometrial biopsy** for histology/GeneXpert
PCOS	Oligo/amenorrhoea, hirsutism, acne, obesity, acanthosis	Ultrasound (≥20 follicles per ovary), testosterone, LH:FSH, AMH; exclude others
Hyperprolactinaemia	Galactorrhoea, amenorrhoea, drugs, headache	**Prolactin** (repeat), TSH, MRI pituitary
Thyroid disease	Symptoms/signs of hypo-/hyperthyroidism, menstrual change	**TSH**, free T4
Hypothalamic amenorrhoea / Sheehan's	Low BMI, exercise, stress; or failure of lactation after severe PPH	Low FSH/LH/E2; pituitary function tests
Premature ovarian insufficiency	Amenorrhoea <40, hot flushes	**FSH >25 IU/L** (repeat), low E2, low AMH
Asherman's syndrome	Hypomenorrhoea/amenorrhoea after D&C or sepsis; cyclical pain	HSG filling defects; **hysteroscopy**
Uterine fibroids / polyps	Heavy periods, enlarged irregular uterus	**Ultrasound**, saline sonohysterography, hysteroscopy
Endometriosis	Progressive dysmenorrhoea, deep dyspareunia, fixed retroverted uterus, nodules	Ultrasound (endometrioma); **laparoscopy**
Male factor	Abnormal semen analysis, abnormal testicular exam	**Semen analysis** ×2, FSH/LH/testosterone, scrotal US, karyotype
Coital dysfunction	Vaginismus, erectile dysfunction, infrequent intercourse	History (sensitively), examination
Unexplained	All tests normal	Diagnosis of exclusion

8. Clinical Workup

Infertility is an **outpatient, couple-based** workup spread over one or two menstrual cycles. The skill is to order the tests **in the right sequence and at the right time of the cycle**, so that within about 2–3 months you can answer the four questions (eggs, tubes, womb, sperm) and not waste her fertile time. A good first visit ends with a **semen analysis request, a correctly timed progesterone, an HSG date, and basic screening for both partners**.

First 5 minutes (not usually an emergency, but check before a long history):

1. **Is she pregnant now?** Do a **urine pregnancy test** if her period is late or cycles are irregular. Positive with pain or bleeding → ectopic pathway (section 9).
2. **Is she on (or just off) ovulation induction or IVF?** Distension, pain, breathlessness, vomiting or reduced urine → **OHSS**.
3. **Neuro-endocrine red flags**: galactorrhoea with headache or visual field loss; rapid virilisation.
4. **Safety**: ask the woman, **alone**, whether she is safe at home; infertility-related **intimate partner violence** and **depression/suicidal thoughts** are common and change today's plan.
5. **Baseline vitals and BMI** for both partners (BMI under 19 or over 30 changes the plan: weight change can restore ovulation, and many units ask for weight loss before ovulation induction when BMI is well above 30, though cut-offs vary).

RED FLAG

Same-day senior review for: a positive pregnancy test with pain or bleeding (ectopic), suspected severe OHSS, visual field loss with galactorrhoea, or disclosure of violence or suicidal intent.

Availability reflects the source notes Verify *compiled from the source notes, not an MoH list*.

Bedside

Test	What it tells you	Available at
Urine pregnancy test	Before any HSG, any drug, and in any woman with a late period	All levels
BMI and waist circumference	Weight management target (BMI 19–30); PCOS metabolic risk	All levels
BP	Hypertension (PCOS, pre-pregnancy optimisation)	All levels
Urine dipstick	Glycosuria (diabetes), leucocytes/nitrites (UTI)	All levels
Random blood sugar	Diabetes (PCOS, obesity); both partners if at risk	HC III upwards
Rapid HIV and syphilis tests (both partners)	Safer conception plan; treat syphilis before pregnancy	All levels
HemoCue / Hb	Anaemia (heavy periods, fibroids) to correct before pregnancy or surgery	HC III upwards
VIA (if due)	Cervical screening should be up to date before fertility treatment	HC III upwards
Transvaginal ultrasound in clinic (where available)	Fibroids and their relation to the cavity, polycystic ovaries (**≥20 follicles per ovary** or volume **≥10 mL**), endometrioma, **hydrosalpinx** (tubular fluid-filled adnexal structure), and **follicle tracking** during ovulation induction (a leading follicle of about 18 mm or more is mature)	HC IV (basic) upwards

Basic lab

Test	When	Expected / abnormal finding	How it changes management (and where)
Semen analysis (2–7 days' abstinence, delivered within 1 hour at body temperature)	**First test for every couple**, before any invasive test on the woman	WHO 2021 lower reference limits (table below)	Normal → focus on the woman. Abnormal → **repeat after about 3 months** (sooner if azoospermic/severe), male hormones, lifestyle; severe → IUI/IVF/ICSI. Most general/regional hospital labs and private labs
Mid-luteal progesterone	Regular cycles; **7 days before the expected period** (day 21 of 28, day 28 of 35)	**>30 nmol/L** confirms ovulation. Lower: check timing and repeat	Ovulatory → look at tubes and semen; anovulatory → hormone workup and ovulation induction. Regional/national referral and private labs

Test	When	Expected / abnormal finding	How it changes management (and where)
FSH, LH, oestradiol (days 2–5)	Oligo/amenorrhoea; age ≥35 (reserve)	High FSH (**>25 IU/L** repeated, under 40) = POI (WHO III); low FSH/LH/E2 = WHO I; normal FSH with high LH:FSH = PCOS pattern	Chooses treatment: letrozole/clomiphene only work in WHO II; donor eggs for POI; gonadotrophins for WHO I. Regional/private labs
Prolactin, TSH	Oligo/amenorrhoea, galactorrhoea, before ovulation induction	Raised prolactin (repeat, unstressed); abnormal TSH	Cabergoline/bromocriptine, MRI pituitary; levothyroxine. Regional/private
Testosterone (± SHBG)	Hirsutism, suspected PCOS	Mildly raised = PCOS; **very high** or rapid virilisation = tumour	Tumour workup; supports Rotterdam criterion
HIV, syphilis, hepatitis B; Hb, blood group, sickle status (both as relevant)	All	Positive results	ART/PrEP and safer conception; treat syphilis; preconception care
Glucose/HbA1c, lipids	PCOS, obesity	Diabetes, dyslipidaemia	Metabolic care before pregnancy
Cervical screening	If due	VIA/HPV positive	Treat before fertility treatment

Semen analysis: how to collect by **masturbation** into a clean wide-mouthed container after **2–7 days of abstinence**; deliver to the lab **within 1 hour**, kept at body temperature. If abnormal, **repeat after about 3 months** (a spermatogenic cycle), sooner if azoospermic or severely abnormal.

Parameter (WHO 2021, 6th edition)	Lower reference limit (5th centile)
Semen volume	**1.4 mL**
Sperm concentration	**16 million/mL**
Total sperm number	**39 million per ejaculate**
Total motility (progressive + non-progressive)	**42%**
Progressive motility	**30%**
Vitality (live sperm)	**54%**
Normal forms (morphology)	**4%**

> NOTE
> These are the **5th centile of fertile men**, not a pass/fail mark: values slightly below them lower the chance of conception but do not mean sterility, and values above them do not guarantee fertility.

Imaging

Test	When	Expected / abnormal finding	How it changes management (and where)
Pelvic ultrasound (TVUS)	All women	Fibroids, polyps, endometrioma, hydrosalpinx, polycystic ovaries	Myomectomy/hysteroscopy before treatment; salpingectomy for hydrosalpinx before IVF. HC IV (basic) upwards

Test	When	Expected / abnormal finding	How it changes management (and where)
HSG	Tubal assessment in all (after normal semen analysis); **days 6–10** of the cycle, pregnancy excluded	Free bilateral spill = patent; block, hydrosalpinx, loculated spill, beaded tubes, cavity defects	Interpretation in "Putting it together". Screen for chlamydia or give **antibiotic prophylaxis (e.g. doxycycline)**; not with active pelvic infection. Regional/national referral and private radiology
HyCoSy / saline sonohysterography	Where available, instead of HSG or to define cavity lesions	Tubal spill, polyps, submucous fibroids	No radiation. Mainly private/specialist centres
Scrotal ultrasound (man)	Abnormal semen analysis, suspected varicocele or mass	Varicocele, testicular mass, obstruction	Urology. Hospitals
MRI pituitary	High prolactin, visual field loss	Micro- or macroadenoma	Dopamine agonist; neurosurgery. NRH and private

Specialised / referral-level

Test	When	Expected / abnormal finding	How it changes management (and where)
AMH / antral follicle count	Age ≥35, before IVF, suspected low reserve or PCOS	Low = fewer eggs; high = PCOS	Predicts response to stimulation (not natural fertility); urgency of referral. AMH mainly private labs in Kampala
Laparoscopy and dye	History of PID, endometriosis or pelvic surgery, abnormal/equivocal HSG	Tubal patency, peri-tubal adhesions, endometriosis	**Gold standard**; diagnose and treat at once. Few national referral and private centres
Hysteroscopy	Suspected Asherman's, polyp, submucous fibroid, septum	Adhesions, lesions	Diagnose **and** treat. Limited centres
Endometrial biopsy (histology + GeneXpert)	Suspected genital TB	Granulomas, MTB detected	Anti-TB therapy (2RHZE/4RH); tubes rarely recover
Male FSH, LH, testosterone	Low count/azoospermia	High FSH → testicular failure; low FSH/LH/testosterone → hypogonadotropic hypogonadism; normal FSH + azoospermia → obstruction likely	Obstructive vs non-obstructive azoospermia; ICSI, sperm retrieval or donor sperm
Post-ejaculatory urine	Low-volume ejaculate	Sperm in urine	Retrograde ejaculation
Karyotype, Y-chromosome microdeletions	Severe oligozoospermia (<5 million/mL) or non-obstructive azoospermia	Klinefelter 47,XXY, microdeletions	Counselling; micro-TESE or donor sperm
CFTR testing	Congenital bilateral absence of the vas	CF mutations	Partner testing; sperm retrieval + ICSI

Test	When	Expected / abnormal finding	How it changes management (and where)
Urethral swab/urine NAAT	STI; leucocytes in semen	Positive	Treat both partners

UGANDA CONTEXT

At **HC III/IV** you can take the history, examine both partners, do a pregnancy test, rapid HIV/syphilis tests, Hb, VIA and (sometimes) an ultrasound, start folic acid and lifestyle advice, treat STIs in both, and **refer with a semen analysis request**. Semen analysis, hormones and **HSG** are at **general/regional referral hospitals and private facilities**; laparoscopy, hysteroscopy, AMH, IUI and IVF are at **national referral and private fertility centres** (public IVF at Mulago Specialised Women and Neonatal Hospital since 2025). Costs are paid out of pocket: order tests in a sensible sequence and avoid repeats.

Putting it together

Model one-line summary: "Mr and Mrs K: she is 31, P1+1, with 3 years of secondary infertility after a septic incomplete abortion treated by D&C; regular 28-day cycles; bimanual shows a fixed retroverted uterus. He is 36, treated twice for urethral discharge; normal testes and vasa. Semen analysis normal; day-21 progesterone 42 nmol/L; HSG shows bilateral hydrosalpinges with no spill."

Problem list

1. Secondary infertility, 3 years.
2. **Tubal factor**: bilateral distal tubal occlusion with hydrosalpinges (post-infective).
3. Ovulating (progesterone 42 nmol/L); semen normal.
4. Psychosocial and financial burden.

Working diagnosis: secondary infertility due to post-infective bilateral distal tubal disease. **Plan**: counsel the couple together; realistic options are **IVF** (with **laparoscopic salpingectomy or proximal tubal occlusion first**, because hydrosalpinx fluid halves IVF success), adoption or acceptance; ovulation induction would be useless; warn about ectopic risk if she conceives; folic acid; treat both for any STI.

EXAM FAVOURITE

How to present an infertile couple: "Mr and Mrs O, married 4 years, have been trying to conceive for 3 years: secondary infertility. Mrs O is 29, P1+1; her last pregnancy 4 years ago ended in an incomplete abortion treated by D&C at a private clinic, complicated by fever and lower abdominal pain for 2 weeks. Her cycles are regular, 28 days, normal flow. Mr O is 35, has no children, was treated for urethral discharge twice. Examination of Mrs O: BMI 24, no hirsutism, bimanual shows a fixed retroverted uterus with bilateral adnexal thickening. Mr O: normal testes and vasa. My impression is secondary infertility, most likely tubal factor from post-abortal sepsis; differentials include Asherman's syndrome and male factor. Plan: semen analysis, mid-luteal progesterone, HSG, HIV/syphilis tests for both, counsel together."

Model summary line (PCOS case): "Mrs A, 26, nulliparous, 2 years of primary infertility with oligomenorrhoea (cycles every 45–90 days), hirsutism and BMI 32; her husband's semen analysis is normal and HSG shows bilateral free spill. Pelvic ultrasound shows 24 follicles in each ovary; prolactin and TSH are normal. Impression: primary infertility due to anovulation from PCOS (WHO group II). Plan: lifestyle and weight loss, folic acid, letrozole 2.5 mg days 3–7 with mid-luteal progesterone (timed to her cycle) and follicle tracking, increase to 5 mg if no ovulation; screen for diabetes; review in one cycle."

Worked example 1: interpreting a "day-21" progesterone

Scenario	Result	Interpretation	Next step
Regular 28-day cycle, sample on day 21, period came on day 28	**45 nmol/L**	Ovulation confirmed (>30)	Move on to tubes and semen
Regular **35-day** cycle, sample taken on day 21	**8 nmol/L**	**Mistimed**: day 21 of a 35-day cycle is around ovulation, before the luteal peak	**Repeat on day 28** (7 days before the expected period); e.g. 38 nmol/L on day 28 = ovulating
Regular 28-day cycle, correctly timed	**18 nmol/L**	Equivocal: below the >30 nmol/L cut-off. Many labs treat roughly 16–30 nmol/L as "repeat" rather than proof of anovulation (thresholds vary by lab)	Repeat next cycle, confirm timing (was the period 7 days later?) ± follicle tracking
Cycles every 45–90 days	(No point doing "day 21")	Oligo-ovulation is clinically obvious	FSH/LH/E2 days 2–5, prolactin, TSH, testosterone, ultrasound → e.g. PCOS → letrozole after semen and tubes are checked
Amenorrhoea	FSH 48 IU/L on two occasions, age 34	**POI** (WHO III)	Counsel; donor-egg IVF; HRT for bone and heart

Worked example 2: reading an HSG report

HSG finding	What it means	Next step
Normal triangular cavity, **free bilateral peritoneal spill**	Tubes patent	Tubal factor unlikely (HSG can miss adhesions/endometriosis)
Bilateral cornual (proximal) block, no tubes seen	True block **or spasm**/debris	Confirm with **laparoscopy and dye** (or repeat) before labelling; tubal cannulation or IVF
Bilateral dilated, club-shaped tubes with no spill	**Bilateral hydrosalpinges** (distal occlusion, post-PID)	**IVF**, with **salpingectomy or proximal occlusion first**; surgery (salpingostomy) only for mild disease in younger women
One tube patent with spill, the other blocked	At least one patent tube	Natural conception or ovulation induction/IUI possible if needed
Spill present but **loculated** (contrast pooling around the tube)	**Peri-tubal adhesions**	Laparoscopy (adhesiolysis)
Beaded, rigid "pipe-stem" tubes, calcification, irregular small cavity	**Genital TB**	Endometrial biopsy (histology, GeneXpert), chest X-ray; anti-TB therapy; IVF only if the endometrium is healthy
Filling defect in the cavity	Submucous fibroid or polyp	Ultrasound/hysteroscopy → resection
Irregular cavity with multiple filling defects, poor filling	**Asherman's** (intrauterine adhesions)	Hysteroscopic adhesiolysis + oestrogen

Worked example 3: reading a semen analysis against WHO 2021

> Mr O, 38, 3 days' abstinence, delivered in 40 minutes: volume **2.8 mL**; concentration **9 million/mL**; total **25 million**; total motility **35%**; progressive motility **20%**; vitality **62%**; normal forms **2%**.

Parameter	Result	Lower reference limit	Verdict
Volume	2.8 mL	1.4 mL	Normal

Parameter	Result	Lower reference limit	Verdict
Concentration	9 million/mL	16 million/mL	**Low (oligozoospermia)**
Total number	25 million	39 million	**Low**
Total motility	35%	42%	**Low**
Progressive motility	20%	30%	**Low (asthenozoospermia)**
Vitality	62%	54%	Normal
Normal forms	2%	4%	**Low (teratozoospermia)**

- **Interpretation**: **oligo-astheno-teratozoospermia (OAT)**, moderate (not severe: concentration is above 5 million/mL, so karyotype is not yet indicated).
- **Next step**: history and examination of him (steroids, heat, smoking, alcohol, varicocele, testicular volume), treat any infection, lifestyle changes, **repeat the analysis after about 3 months** (spermatogenesis takes about 74 days); **FSH, LH, testosterone** if it stays abnormal. If it persists: **IUI** with washed sperm is possible for mild male factor; moderate–severe OAT → **IVF/ICSI**. Remember these limits are the **5th centile of fertile men**, not a pass/fail mark.
- **Contrast**: **azoospermia with low volume (<1.4 mL)** and **impalpable vasa** → congenital bilateral absence of the vas (CFTR testing; sperm retrieval + ICSI). Azoospermia with **small testes and high FSH** → non-obstructive (karyotype; micro-TESE or donor sperm).

9. Management

Immediate / emergency

Infertility itself is never an emergency, but three emergencies present in the fertility clinic:

- **Ectopic pregnancy**: positive pregnancy test with pain or bleeding (damaged tubes, after tubal surgery or IVF). **ABCDE**: airway, breathing, pulse, BP and shock index; IV access with two large-bore cannulae, group and cross-match, Hb; ultrasound; senior review today; resuscitate and take to theatre if unstable.
- **Severe OHSS** (after ovulation induction or IVF; management below): ABCDE, admit.
- **Intimate partner violence or suicidal intent**: safety assessment the same day, counselling, and referral to support services.

Ovarian hyperstimulation syndrome (OHSS)

Why it happens: many follicles + hCG → excessive **VEGF** → leaky capillaries → fluid shifts into the abdomen (ascites) and chest; the patient is intravascularly **dry** but "waterlogged" → haemoconcentration, oliguria, **thrombosis**.

1. **Assess**: abdominal distension/pain, nausea, breathlessness, urine output, weight, girth; **Hct** (>45% severe), electrolytes, renal and liver function, ultrasound (ovarian size, ascites).
2. **Mild–moderate**: outpatient: oral fluids to thirst, paracetamol, **avoid NSAIDs**, daily weight, review.
3. **Severe (admit)**: strict **fluid balance**; IV crystalloid to maintain urine output; **LMWH thromboprophylaxis** (e.g. enoxaparin 40 mg SC daily); **paracentesis** for tense ascites or respiratory compromise; albumin in selected cases; avoid pelvic examination (ovaries rupture/torsion).
4. **Prevention**: lowest effective doses, ultrasound monitoring, cancel or "freeze-all" when too many follicles, GnRH-agonist trigger in IVF.

Definitive

1. General measures for every couple (HC IV and above)

1. **Counsel together**: explain the normal chance of conception (about 20% per month; 80% in a year), that infertility is a shared problem, and outline the plan and costs.
2. **Intercourse every 2–3 days** throughout the cycle.
3. **Folic acid 400 micrograms PO daily** for the woman from now until 12 weeks of pregnancy (**5 mg daily** if BMI ≥30, diabetes, epilepsy on anticonvulsants, sickle cell disease or previous neural tube defect; in Uganda 5 mg tablets are what is usually stocked).
4. **Weight**: aim for BMI 19–30. A **5–10% weight loss** restores ovulation in many women with PCOS; underweight women need to gain weight.
5. **Stop smoking**, minimise alcohol, avoid recreational drugs and **anabolic steroids/testosterone** (men); avoid excessive heat to the scrotum.
6. **Treat infections in both partners** (syndromic STI treatment; TB treatment if found).
7. **Review medications** (antipsychotics, metoclopramide) and chronic diseases (diabetes, thyroid, hypertension, HIV).
8. **HIV**: test both. For **serodiscordant couples**: the positive partner on ART with a **suppressed viral load** does not transmit HIV sexually (**U = U**); offer **PrEP** to the negative partner; time condomless intercourse to the fertile window; sperm washing where available.
9. **Cervical screening**, rubella/hepatitis B status where available, and pre-conception care.
10. **Psychological support**; ask about violence; link to counselling/support groups.

2. Ovulation induction (anovulation: WHO group II, mostly PCOS)

Before starting: confirm anovulation, **normal semen analysis**, **at least one patent tube** (HSG), exclude pregnancy, treat thyroid/prolactin problems, optimise weight. Drugs and doses: drug table.

How the drugs work (the "why")

- **Clomiphene** blocks **oestrogen receptors in the hypothalamus**, so the brain "thinks" oestrogen is low → more GnRH → more **FSH** → follicle growth. Its long half-life also blocks oestrogen receptors in the endometrium and cervix (thin lining, hostile mucus): part of why pregnancy rates lag behind ovulation rates.
- **Letrozole** blocks **aromatase** → less oestrogen made → removes negative feedback → more FSH. Its short half-life means the endometrium is spared and **mono-follicular** ovulation is more likely.
- Both need an **intact hypothalamic–pituitary axis**, which is why they **do not work in WHO group I** (low gonadotrophins) or **group III** (failed ovary).

Monitoring and when to escalate

1. Confirm ovulation each cycle with **mid-luteal progesterone** (± follicle tracking).
2. **Clomiphene resistance** = no ovulation at 150 mg → add metformin, switch to letrozole, or move to gonadotrophins or **laparoscopic ovarian drilling** (LOD: diathermy of each ovary; restores ovulation without multiple-pregnancy risk; risk of adhesions).
3. **Clomiphene failure** = ovulating but no pregnancy after 6 cycles → re-evaluate (tubes, semen) and consider gonadotrophins ± IUI or IVF.

Other causes of anovulation

Cause	Treatment
Hyperprolactinaemia	Stop causative drugs; treat hypothyroidism; **cabergoline** or **bromocriptine** (drug table); MRI pituitary; usually stop once pregnant
Hypothyroidism	Levothyroxine 25–50 micrograms daily, titrated to TSH

Cause	Treatment
WHO group I (hypothalamic)	Weight gain, reduce exercise; **gonadotrophins with LH activity (hMG)** or pulsatile GnRH (specialist)
POI (WHO III)	**Egg donation IVF**; HRT until the age of natural menopause for bone and heart

3. Tubal and peritoneal disease

Situation	Management
Mild distal disease / peri-tubal adhesions	**Laparoscopic adhesiolysis**, fimbrioplasty or salpingostomy (referral centre), then try naturally for 6–12 months
Proximal (cornual) block	Often spasm or debris: repeat test; **tubal cannulation** (hysteroscopic/fluoroscopic)
Severe/bilateral damage, thick-walled hydrosalpinges, failed surgery, older woman	**IVF**: surgery has poor results
Hydrosalpinx before IVF	**Laparoscopic salpingectomy** (or proximal tubal occlusion): **doubles IVF success**
Genital TB	Anti-TB therapy (2 months RHZE + 4 months RH); tubes rarely recover → IVF (only if endometrium is healthy)
Previous sterilisation	Tubal reanastomosis (microsurgery) or IVF

UGANDA CONTEXT

Open "tuboplasty" by laparotomy is still offered in some Ugandan facilities, often with poor results and new adhesions. Counsel honestly: with severe tubal damage the realistic options are **IVF** (private centres in Kampala; public IVF at Mulago Specialised Women and Neonatal Hospital since 2025), **adoption/fostering**, or acceptance. Ectopic pregnancy risk is high after any tubal surgery: tell her to present early in any pregnancy for an ultrasound.

4. Uterine factors

Problem	Management
Submucous fibroid / fibroid distorting the cavity	**Hysteroscopic resection** (submucous) or **myomectomy** (large intramural): improves pregnancy rates; subserous fibroids usually leave alone
Endometrial polyp	Hysteroscopic polypectomy
Asherman's syndrome	**Hysteroscopic adhesiolysis**, then prevent re-adhesion (intrauterine balloon/IUD for a short time) + **oestrogen** to regrow endometrium; second-look hysteroscopy
Uterine septum	Hysteroscopic resection (recurrent loss/infertility)

5. Endometriosis

- Minimal–mild: **laparoscopic excision/ablation** of lesions and adhesiolysis improves spontaneous pregnancy rates.
- Endometrioma: surgery if large/symptomatic (beware reducing ovarian reserve).
- **Do not** use hormonal suppression (progestogens, COC) as a fertility treatment: it prevents pregnancy while in use.
- Moderate–severe or older women → IVF.

6. Male factor

Finding	Management
Lifestyle / exposure	Stop smoking, alcohol, **anabolic steroids/testosterone**, reduce heat; weight loss; repeat semen analysis after 3 months
Infection (leucocytes, STI)	Treat the man and his partner
Hypogonadotropic hypogonadism	**hCG ± FSH** injections (specialist): highly effective; **never testosterone** (suppresses spermatogenesis)
Mild male factor	IUI (with washed sperm) or IVF
Severe oligo-/astheno-/teratozoospermia	**IVF with ICSI** (single sperm injected into each egg)
Obstructive azoospermia (normal FSH, normal testes)	Surgical sperm retrieval (**PESA/TESA**) + ICSI, or microsurgical reconstruction
Non-obstructive azoospermia (high FSH, small testes)	Karyotype; micro-TESE + ICSI (specialist) or **donor sperm**; adoption
Retrograde ejaculation	Sympathomimetics, or sperm from alkalinised post-ejaculatory urine for IUI/ICSI
Erectile/ejaculatory dysfunction	Treat the cause; PDE5 inhibitors (e.g. sildenafil); psychosexual counselling
Varicocele	NICE: **not** a fertility treatment on its own; repair may be offered for a clinical varicocele with abnormal semen (specialist decision)

7. Unexplained infertility

- Reassure: many conceive naturally. **NICE** advises continuing to try for a total of **2 years**, then **IVF**; it does **not** recommend oral ovulation induction alone for unexplained infertility.
- Where IVF is unaffordable (most of Uganda), **ovarian stimulation + IUI** is often used as a cheaper intermediate step: counsel about multiple pregnancy.
- Age >35 → move to IVF sooner.

8. Assisted reproduction (ART)

Technique	What happens	Indications	Key risks
IUI (intrauterine insemination)	Washed, concentrated sperm placed in the uterus near ovulation (natural or stimulated cycle)	Mild male factor, cervical factor, coital difficulty, serodiscordant couples (HIV-positive man: sperm washing), unexplained (with stimulation)	Multiple pregnancy (stimulated), infection; needs ≥1 patent tube
IVF	Ovarian stimulation → oocyte retrieval (transvaginal needle) → fertilisation in the lab → embryo transfer	Tubal disease, failed other treatments, endometriosis, unexplained, older women	**OHSS**, multiple pregnancy (limit embryos transferred), ectopic, cost, emotional burden
ICSI	Single sperm injected into each oocyte	Severe male factor, surgically retrieved sperm, previous fertilisation failure	As IVF
Donor gametes / surrogacy	Donor eggs (POI), donor sperm	Absent gametes	Legal, cultural and ethical issues

9. Counselling and social aspects

- Always aim to **see the couple together** at least once; explain results jointly: this reduces blame of the woman.
- **Address myths** (e.g. that infertility is always the woman's fault, or caused by past contraceptive use: modern methods do not cause infertility, though DMPA delays return of fertility for some months).
- Screen for **depression, anxiety and intimate partner violence**; involve counsellors, religious leaders or support groups if the couple wishes.
- Discuss **costs and realistic success rates** before expensive treatments; warn against unregulated herbal remedies.
- Discuss **adoption and fostering** (legal adoption in Uganda goes through the courts) as positive options.
- Protect **confidentiality**: a male-factor or HIV result is disclosed only with consent.

Drug table

Drug	Dose (adult)	Route	Frequency	Duration	Key cautions
Folic acid	**400 micrograms** (5 mg if BMI ≥30, diabetes, epilepsy on anticonvulsants, sickle cell, previous NTD)	Oral	Once daily	From now until 12 weeks of pregnancy	5 mg tablets are what Uganda usually stocks
Letrozole (aromatase inhibitor), **first line in PCOS**	**2.5 mg**; increase by 2.5 mg per cycle to max **7.5 mg** if no ovulation	Oral	Once daily on **days 3–7** (or 2–6) of a spontaneous or progestogen-induced bleed	Per cycle	Higher ovulation and live-birth rates than clomiphene; fewer multiples; **off-label**: counsel; exclude pregnancy before each course
Clomiphene citrate (SERM)	**50 mg**; increase by 50 mg per cycle to max **150 mg**	Oral	Once daily on **days 2–6**	**No more than 6 months** (ovulatory cycles; NICE)	**Ultrasound follicle tracking at least in the first cycle**; hot flushes, mood change, **visual disturbance (stop)**, multiple pregnancy (about 5–10% twins), thin endometrium; possible small ovarian cancer association with prolonged use
Metformin	500 mg daily with food, increasing weekly to **500 mg 8-hourly** (1.5 g/day) or 850 mg 12-hourly	Oral	Continuous		Add in **clomiphene-resistant** PCOS or obese women; helps weight and metabolic health; GI side effects
Gonadotrophins (recombinant FSH / hMG)	Low-dose step-up (e.g. 37.5–75 IU/day)	SC/IM	Daily	With **ultrasound monitoring**	Second line (specialist); **multiple pregnancy and OHSS**: cancel if too many follicles
hCG trigger	5,000–10,000 IU	IM	Single dose	When the leading follicle is about 18 mm or more	Times ovulation/IUI; OHSS risk

Drug	Dose (adult)	Route	Frequency	Duration	Key cautions
Cabergoline	**0.25 mg** twice weekly (0.5 mg/week), increasing monthly by 0.5 mg/week according to prolactin	Oral	Twice weekly	Usually stop once pregnant	
Bromocriptine	**1.25 mg** at night with food, increasing to 2.5 mg 8–12-hourly	Oral	Nightly, then 8–12-hourly	Usually stop once pregnant	Nausea, postural hypotension
Levothyroxine	25–50 micrograms, titrated to TSH	Oral	Once daily	Long term	
Doxycycline (HSG prophylaxis, e.g.)	Per unit protocol	Oral		Around the HSG	Exclude pregnancy first; not with active pelvic infection
Anti-TB therapy (genital TB)	**2 months RHZE + 4 months RH** (standard weight-band dosing)	Oral	Daily	6 months	Tubes rarely recover
Enoxaparin (severe OHSS)	**40 mg**	SC	Once daily	While at risk	Avoid NSAIDs in OHSS
Paracetamol (OHSS)	1 g	Oral	6-hourly		Max 4 g/day
hCG ± FSH (male hypogonadotropic hypogonadism)	Specialist dosing	SC/IM		Months	**Never testosterone**

Referral criteria

Level	What you can do	Refer up when
HC III/IV	Take a full couple history, examine both, counsel, lifestyle and folic acid, treat STIs in both partners, HIV testing/safer conception, pregnancy test, Hb, VIA, request semen analysis where a lab exists	Every couple meeting the definition (or at 6 months if she is ≥35) → General Hospital / RRH with a semen analysis request; same day if ectopic or OHSS suspected
General / Regional Referral Hospital	Semen analysis, mid-luteal progesterone, hormones (FSH, LH, prolactin, TSH), pelvic ultrasound, **HSG**, **ovulation induction with letrozole/clomiphene with ultrasound tracking**, myomectomy, some diagnostic laparoscopy	Tubal disease needing surgery or IVF, azoospermia or severe male factor, POI, clomiphene/letrozole failure, Asherman's, endometriosis, pituitary adenoma → NRH / fertility centre
National referral / fertility centres	Laparoscopy and hysteroscopy, tubal surgery, gonadotrophins, IUI, **IVF/ICSI**, surgical sperm retrieval, gamete donation	n/a

Monitoring and follow-up

- **Ovulation induction**: mid-luteal progesterone every cycle (timed 7 days before the expected period) ± follicle tracking (at least the first clomiphene cycle); pregnancy test before each course; review after each cycle; stop clomiphene after 6 ovulatory cycles.
- **Male factor**: repeat semen analysis after about 3 months of lifestyle change or treatment.
- **After tubal surgery or IVF**: early ultrasound in any pregnancy (ectopic risk).
- **Unexplained**: review at a total of 2 years of trying (sooner if over 35).
- **Psychosocial**: depression and violence at every visit.

10. Complications

Disease-related

- Marital breakdown, stigma, violence, depression.
- Financial ruin from repeated ineffective treatment.
- Ectopic pregnancy when conception does occur (damaged tubes).
- Long-term PCOS risks: diabetes, endometrial hyperplasia/cancer, cardiovascular disease.
- Consequences of the cause: chronic pelvic pain (PID), osteoporosis (POI).

Treatment-related

Complication	Cause	Recognition and action
Multiple pregnancy → preterm birth, pre-eclampsia, PPH	Clomiphene (about 5–10% twins); gonadotrophins and IVF higher	Ultrasound monitoring; cancel cycles with too many follicles; limit embryos transferred
OHSS: ascites, thrombosis, renal failure	Gonadotrophins, hCG	Section 9
Ectopic pregnancy	Tubal surgery, IVF	Early ultrasound
Surgical complications: adhesions, bleeding	Laparoscopy, myomectomy, ovarian drilling	Counsel; skilled surgeon
Visual disturbance, hot flushes; possible small ovarian cancer association with prolonged use	Clomiphene	Stop if visual symptoms; limit to 6 months
Reduced ovarian reserve	Endometrioma surgery	Careful case selection

11. Prognosis

- **Age** is the single strongest predictor of success; female fertility falls from the early 30s and steeply after 37.
- About 80% of couples conceive within 1 year and 90% within 2 years; many with unexplained infertility conceive naturally.
- **Anovulation (PCOS)** responds well to weight loss and letrozole; **POI** needs donor eggs.
- **Severe tubal damage** has a poor prognosis without IVF; hydrosalpinx halves IVF success unless removed; **genital TB** rarely allows tubal recovery.
- Semen values just below the WHO limits lower, but do not abolish, the chance of conception.

12. Prevention

- **Prevent tubal infertility**: early syndromic treatment of STIs **for both partners**; early treatment of PID; **safe post-abortion care** (MVA and antibiotics, avoid sharp curettage of a soft uterus); **clean deliveries and caesarean sections with prophylactic antibiotics**; TB control.
- **Male**: mumps prevention, prompt treatment of urethritis and epididymo-orchitis, avoid anabolic steroids, smoking, heavy alcohol and scrotal heat.
- **Lifestyle**: healthy weight (BMI 19–30), no smoking.
- **Do not waste fertile time**: investigate at 12 months (6 months if ≥35), semen and tubes before ovulation induction.
- **Contraception myths**: modern methods do not cause infertility (DMPA delays return for some months).

13. Special Populations

- **Women aged 35 or over**: investigate at 6 months; check ovarian reserve (FSH, AMH, antral follicle count); refer early for ART rather than months of oral drugs.
- **HIV serodiscordant couples**: positive partner on ART with suppressed viral load (U = U), PrEP for the negative partner, timed condomless intercourse in the fertile window, sperm washing/IUI where available.
- **Obesity (BMI >30) and PCOS**: 5–10% weight loss; metformin; many units ask for weight loss before ovulation induction.
- **Underweight / hypothalamic amenorrhoea**: weight gain; letrozole and clomiphene will not work.
- **After severe PPH (Sheehan's)**: pituitary function tests; gonadotrophins.
- **Sickle cell disease**: folic acid 5 mg; hydroxyurea affects sperm and must be planned with haematology Verify *hydroxyurea and male fertility advice not in the source notes*.
- **Previous sterilisation**: tubal reanastomosis or IVF.

14. Key Points

High-yield points

- Infertility is a disease of the couple: examine and test the man.
- Four questions: eggs, tubes, womb, sperm (plus "is intercourse happening?").
- Semen analysis first; HSG on days 6–10.
- Mid-luteal progesterone 7 days before the expected period; >30 nmol/L.
- Oligo/amenorrhoea needs hormones, not a progesterone.
- In Uganda, secondary tubal infertility from infection dominates; prevention is the best treatment.
- Letrozole is first-line in PCOS; never induce ovulation before semen and tubes are checked.
- Hydrosalpinx halves IVF success: remove or clip it first.
- Never testosterone for a man with a low count.
- Counsel together; screen for depression and violence.

Numbers & doses

Item	Value
Definition of infertility	No clinical pregnancy after **≥12 months** of regular unprotected intercourse (investigate at **6 months** if she is **≥35** or there is an obvious cause)

Item	Value
Chance of conception	About **20%** per cycle; **80%** in 1 year; **90%** in 2 years
Fertile window	The **5 days before** ovulation **and the day of** ovulation; advise intercourse **every 2–3 days**
LH surge to ovulation	About **36 hours**
Luteal phase	Fixed at about **14 days**
Mid-luteal progesterone	**7 days before the expected period**; **>30 nmol/L** confirms ovulation
HSG timing	**Days 6–10** (after the flow, before the egg), pregnancy excluded, antibiotic prophylaxis or chlamydia screen
Semen collection	**2–7 days** abstinence; to the lab **within 1 hour** at body temperature; repeat abnormal after about **3 months**
Spermatogenesis	About **74 days** (about 3 months including transport)
WHO 2021 semen limits	Volume **1.4 mL**; concentration **16 million/mL**; total **39 million**; total motility **42%**; progressive **30%**; vitality **54%**; normal forms **4%**
Karyotype/Y-microdeletion	Concentration **<5 million/mL** or non-obstructive azoospermia
Normal testicular volume	About **15–25 mL**
POI	Amenorrhoea **<40 years** with **FSH >25 IU/L** (repeated)
PCOS ultrasound	**≥20 follicles per ovary** or ovarian volume **≥10 mL** (or raised AMH in adults)
Letrozole	**2.5 mg** PO daily **days 3–7** (or 2–6); increase by 2.5 mg per cycle to max **7.5 mg**
Clomiphene	**50 mg** PO daily **days 2–6**; increase to max **150 mg**; ultrasound tracking at least in the first cycle; **≤6 months**; twins about **5–10%**
Metformin (PCOS)	Build up to **500 mg 8-hourly** (1.5 g/day) or 850 mg 12-hourly
hCG trigger	**5,000–10,000 IU IM** when the leading follicle is about **18 mm** or more
Cabergoline	**0.25 mg twice weekly** (0.5 mg/week), titrated monthly
Bromocriptine	**1.25 mg at night** with food, up to 2.5 mg 8–12-hourly
Folic acid	**400 micrograms** daily (**5 mg** if BMI ≥30, diabetes, epilepsy on drugs, sickle cell, previous NTD)
Weight loss in PCOS	**5–10%** often restores ovulation; target BMI **19–30**
DMPA	Return of fertility can take up to about **10 months** after the last injection
Unexplained infertility (NICE)	Try for a total of **2 years**, then IVF
Genital TB	**2 months RHZE + 4 months RH**
OHSS	Severe if haematocrit **>45%**; LMWH e.g. **enoxaparin 40 mg SC daily**; avoid NSAIDs

Exam traps

- **Giving clomiphene "to try" at the first visit.** Wrong: check **semen** and **at least one patent tube** first; ovulation induction with blocked tubes or azoospermia wastes time and risks ectopic and multiple pregnancy.

- **"Day-21 progesterone" in a 35-day cycle.** Wrong: take it **7 days before the expected period** (day 28); a low day-21 value in a long cycle is usually mistimed.
- **Ordering a progesterone in amenorrhoea.** Wrong: she is obviously not ovulating; do **FSH, LH, E2, prolactin, TSH, testosterone** and a pregnancy test.
- **Doing an HSG before a semen analysis.** Wrong: the semen analysis is cheap, non-invasive and should come first.
- **Clomiphene or letrozole for WHO group I or III.** Wrong: both need an intact hypothalamic–pituitary axis and a working ovary; group I needs gonadotrophins (hMG) or pulsatile GnRH, group III needs donor eggs.
- **Testosterone for a man with a low sperm count.** Wrong: it suppresses FSH/LH and switches off spermatogenesis.
- **Labelling bilateral cornual block on HSG as definite tubal occlusion.** Wrong: it may be **spasm**; confirm with laparoscopy and dye.
- **Treating a WHO semen value just below the limit as sterility.** Wrong: the limits are the **5th centile of fertile men**; repeat after 3 months and interpret the whole picture.
- **Hormonal suppression for endometriosis-associated infertility.** Wrong: it prevents pregnancy while in use.

Questions seniors ask

On the ward round / in clinic

- **What four questions does the infertility workup answer?** Is she ovulating? Are the tubes open? Is the cavity normal? Is the semen normal? (And is intercourse actually happening?)
- **When do you take the mid-luteal progesterone, and what confirms ovulation?** Seven days before the expected period (day 21 of 28, day 28 of 35); >30 nmol/L.
- **When is HSG done, and what precautions?** Days 6–10, after excluding pregnancy, with chlamydia screening or antibiotic prophylaxis; never with active pelvic infection.
- **Give the WHO 2021 semen reference limits.** 1.4 mL, 16 million/mL, 39 million total, 42% total motility, 30% progressive, 54% vitality, 4% normal forms.
- **Azoospermia with normal testes and normal FSH: what is likely?** Obstruction (post-infective or CBAVD): check the vasa, then sperm retrieval (PESA/TESA) + ICSI or reconstruction.
- **HSG shows bilateral hydrosalpinges: what do you tell the couple?** Natural conception is very unlikely; the realistic option is IVF after salpingectomy or proximal tubal occlusion (hydrosalpinx halves IVF success); ectopic risk is high if she conceives.
- **Why is letrozole first-line in PCOS?** Higher ovulation and live-birth rates than clomiphene, mostly mono-follicular, no anti-oestrogenic effect on the endometrium (2023 PCOS guideline).
- **What would you do for a 38-year-old with 1 year of infertility?** Investigate both partners now (she qualified at 6 months), include ovarian reserve (FSH/AMH/antral follicle count), and refer early for assisted reproduction rather than months of oral drugs.

Viva questions

1. *Define infertility and distinguish primary from secondary.* No clinical pregnancy after 12 months of regular unprotected intercourse; primary = never conceived, secondary = previous pregnancy.
2. *What are the Rotterdam criteria?* 2 of 3: oligo/anovulation, clinical/biochemical hyperandrogenism, polycystic ovarian morphology (≥20 follicles per ovary or ovarian volume ≥10 mL, or raised AMH in adults), after excluding other causes.
3. *When and why is the "day-21" progesterone done?* 7 days before the expected period, at the peak of corpus luteum progesterone; >30 nmol/L confirms ovulation.
4. *When is HSG done and why then?* Days 6–10: after menstruation (clear view, lower infection spread) and before ovulation (avoid irradiating an early pregnancy).

5. *How does clomiphene work, and what are its risks?* Anti-oestrogen at the hypothalamus → increased FSH; risks multiple pregnancy, OHSS, visual disturbance, thin endometrium; limit to 6 months.
6. *Why letrozole over clomiphene in PCOS?* Higher ovulation and live-birth rates, more mono-follicular, no anti-oestrogenic effect on endometrium.
7. *Give the WHO 2021 semen reference values.* Volume 1.4 mL, concentration 16 million/mL, total 39 million, total motility 42%, progressive 30%, vitality 54%, normal forms 4%.
8. *Why remove a hydrosalpinx before IVF?* The fluid is embryotoxic and washes embryos out; removal roughly doubles IVF success.
9. *Why must a man with low sperm not take testosterone?* Exogenous testosterone suppresses FSH/LH via negative feedback and switches off spermatogenesis.
10. *Why is secondary infertility so common in Uganda?* Tubal damage from STIs/PID, unsafe abortion, post-partum sepsis and TB.

Mnemonics

- **The four pillars: "Eggs, Tubes, Womb, Sperm"** (ovulation, tubal, uterine, male), plus "**Sex**" (is intercourse actually happening?).
- **Mid-luteal progesterone: "7 before"**: 7 days before the next period, not always day 21.
- **HSG: "after the flow, before the egg"** (days 6–10).
- **WHO 2021 semen values: "1.4 – 16 – 39 – 42 – 30 – 54 – 4"**: volume, concentration, total count, total motility, progressive, vitality, morphology.
- **WHO anovulation groups: "I = Idle brain, II = PCOS in between, III = Ovary Out"**.
- **Clomiphene: "50 up to 150, days 2–6, 6 months max, scan the first."**
- **Letrozole: "2.5, 3 to 7, up to 7.5."**
- **Rotterdam: "2 of 3: Oligo, Hairy, Cysty."**

OSCE tips

> EXAM FAVOURITE
> **OSCE: infertility history station:** introduce yourself to both partners, explain confidentiality, take the woman's history (age, duration, obstetric, menstrual, PID/STI, surgery, PCOS/prolactin/thyroid symptoms, drugs, lifestyle), the man's history (children, mumps, testes, STIs, surgery, sexual function, drugs/steroids, heat), coital frequency, HIV status, and the **psychosocial impact**; summarise and outline the four-part investigation plan (ovulation, tubes, uterus, semen).

- **Results station**: read a progesterone against cycle length (worked example 1), an HSG report (worked example 2) and a semen analysis against WHO 2021 limits (worked example 3).
- **Counselling station**: explain results to the couple together, address blame and myths, give realistic options and costs, mention adoption and fostering, and check for violence.

15. References

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SELF-TEST

Flashcards

25 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	Define infertility (WHO).	Failure to achieve a **clinical pregnancy after ≥12 months of regular unprotected sexual intercourse**. **Primary**: never conceived. **Secondary**: previous pregnancy of any outcome.
2	When should you investigate before 12 months of trying?	After **6 months** (or at once) if the woman is **≥35**, or there is an obvious risk factor: amenorrhoea/oligomenorrhoea, previous PID/ectopic/pelvic surgery, endometriosis, chemotherapy, known male problem, or HIV-discordant couple planning pregnancy.
3	What proportion of couples conceive within 1 and 2 years?	About **80% within 1 year** and **90% within 2 years**; fecundability about **20% per cycle** in young couples.
4	Why is secondary infertility so common in Uganda?	**Tubal damage** from untreated **gonorrhoea/chlamydia (PID)**, **post-abortal sepsis**, **post-partum/post-caesarean infection** and **genital TB**. HSG studies in Kampala find tubal blockage as the commonest abnormality.
5	The four pillars of infertility evaluation.	1. **Ovulation** (history, mid-luteal progesterone, hormones). 2. **Tubes** (HSG, laparoscopy and dye). 3. **Uterus** (ultrasound, HSG, hysteroscopy). 4. **Semen** (semen analysis). Plus: is intercourse actually happening?
6	When is mid-luteal progesterone taken and what level confirms ovulation?	**7 days before the expected period** (day 21 of a 28-day cycle; day 28 of a 35-day cycle). **>30 nmol/L** confirms ovulation. In irregular cycles repeat weekly until the next period.
7	WHO 2021 semen analysis lower reference limits.	Volume **1.4 mL**; concentration **16 million/mL**; total **39 million**; total motility **42%**; progressive motility **30%**; vitality **54%**; normal forms **4%**. They are 5th centiles of fertile men, not pass marks.
8	How should a semen sample be collected?	Masturbation into a clean wide-mouthed container after **2–7 days of abstinence**; deliver to the lab **within 1 hour** at body temperature. Repeat after **about 3 months** if abnormal (spermatogenesis takes about 74 days).
9	Rotterdam criteria for PCOS (2023 guideline).	**2 of 3**: (1) oligo/anovulation; (2) clinical or biochemical hyperandrogenism; (3) polycystic ovarian morphology — **≥20 follicles per ovary** (or ovarian volume ≥10 mL) on ultrasound, or **raised AMH** in adults — after excluding thyroid disease, hyperprolactinaemia, CAH and androgen-secreting tumours.
10	WHO classification of anovulation.	**Group I**: hypogonadotropic hypogonadism (low FSH/LH/E2 — low weight, Sheehan's, Kallmann). **Group II**: normogonadotropic normo-oestrogenic (**PCOS**). **Group III**: hypergonadotropic hypogonadism (**POI**, high FSH). Hyperprolactinaemia is often listed separately.
11	Letrozole regimen for ovulation induction.	**2.5 mg PO daily on days 3–7**; if no ovulation increase by 2.5 mg per cycle to **max 7.5 mg**. First-line for PCOS (WHO 2025, international PCOS guideline 2023). Off-label — counsel; exclude pregnancy.
12	Clomiphene regimen and safety rules.	**50 mg PO daily on days 2–6**, increasing by 50 mg per cycle to **max 150 mg**; **ultrasound follicle monitoring at least in the first cycle**; **no longer than 6 months** (NICE). Risks: twins (about 5–10%), OHSS, hot flushes, **visual disturbance (stop)**, thin endometrium.

#	Question	Answer
13	How do clomiphene and letrozole induce ovulation?	**Clomiphene**: blocks hypothalamic oestrogen receptors → brain senses low oestrogen → ↑GnRH → ↑FSH. **Letrozole**: blocks aromatase → ↓oestrogen → ↓negative feedback → ↑FSH. Both need an intact hypothalamic–pituitary axis (useless in WHO groups I and III).
14	When is HSG performed and why?	In the **early follicular phase, days 6–10**: after the period has stopped and **before ovulation** (avoids irradiating an early pregnancy). Screen for chlamydia or give **antibiotic prophylaxis**; avoid with active pelvic infection.
15	Why remove a hydrosalpinx before IVF?	Hydrosalpinx fluid refluxes into the cavity, is **embryotoxic** and **washes out embryos** — it about halves IVF success. **Laparoscopic salpingectomy** or proximal tubal occlusion restores success rates.
16	Treatment of hyperprolactinaemia causing anovulation.	Stop causative drugs; exclude hypothyroidism and pregnancy; MRI pituitary; **cabergoline 0.25 mg PO twice weekly** (titrate) or **bromocriptine 1.25 mg at night**, increasing to 2.5 mg 8–12-hourly.
17	Why should an infertile man never be given testosterone?	Exogenous testosterone (and anabolic steroids) **suppress FSH and LH** by negative feedback, which **switches off spermatogenesis** (can cause azoospermia). Hypogonadotropic hypogonadism is treated with **hCG ± FSH**.
18	Azoospermia with normal FSH and normal-sized testes suggests…?	**Obstructive azoospermia** (e.g. post-gonococcal epididymal blockage, absent vas) — sperm production is normal. Options: surgical sperm retrieval (PESA/TESA) + ICSI or reconstruction. High FSH + small testes → non-obstructive (testicular failure).
19	Pathophysiology and key management of OHSS.	Many follicles + hCG → **VEGF** → capillary leak → ascites, pleural effusion, **haemoconcentration**, oliguria, **thrombosis**. Manage: fluid balance, analgesia (no NSAIDs), **LMWH thromboprophylaxis**, paracentesis for tense ascites; prevent with low doses and monitoring.
20	Folic acid dose before conception.	**400 micrograms daily** until 12 weeks; **5 mg daily** if BMI ≥30, diabetes, antiepileptic drugs, sickle cell disease or previous neural tube defect (in Uganda 5 mg tablets are commonly the stocked form).
21	A woman with regular 35-day cycles has a 'day-21' progesterone of 8 nmol/L. Is she anovulatory?	**Not necessarily: the test is mistimed.** Day 21 of a 35-day cycle is around ovulation, before the luteal peak. **Repeat on day 28** (7 days before the expected period); **>30 nmol/L** confirms ovulation.
22	In what order should the basic infertility workup be done?	1. **Semen analysis** (cheap, non-invasive, first). 2. **Ovulation**: mid-luteal progesterone timed to the cycle (or day 2–5 FSH/LH/E2, prolactin, TSH, testosterone if oligo/amenorrhoeic). 3. **Pelvic ultrasound** (cavity, fibroids, hydrosalpinx, ovaries). 4. **Tubes**: **HSG days 6–10** (laparoscopy and dye if history of PID/endometriosis or HSG abnormal). Plus HIV/syphilis/HBV for both. Only then start ovulation induction.
23	HSG shows bilateral dilated club-shaped tubes with no peritoneal spill. Diagnosis and plan?	**Bilateral hydrosalpinges** (distal tubal occlusion, usually post-PID). Natural conception and ovulation induction are futile. Plan: **IVF**, after **laparoscopic salpingectomy or proximal tubal occlusion** (hydrosalpinx fluid halves IVF success); warn of ectopic risk; counsel about adoption/acceptance.
24	Semen analysis: volume 2.8 mL, concentration 9 million/mL, progressive motility 20%, normal forms 2%, vitality 62%. Interpret and plan.	**Oligo-astheno-teratozoospermia** (below WHO 2021 limits of 16 million/mL, 30% progressive, 4% normal forms; volume and vitality normal). Examine him, treat infection, lifestyle (steroids, heat, smoking, alcohol), **repeat after about 3 months**; FSH/LH/testosterone if persistent. Mild → IUI; moderate–severe → **IVF/ICSI**. Karyotype if concentration <5 million/mL.

#	Question	Answer
25	Azoospermia with a semen volume below 1.4 mL and vasa that cannot be felt. Likely cause and next steps?	**Congenital bilateral absence of the vas deferens** (obstructive; linked to CFTR mutations). Normal testes and FSH. Offer **CFTR testing** (both partners), then **surgical sperm retrieval (PESA/TESA) + ICSI**, or donor sperm.

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. A 27-year-old P1+1 has been unable to conceive for 3 years. Her last pregnancy ended in an unsafe abortion followed by 2 weeks of fever and lower abdominal pain treated at a drug shop. Her cycles are regular every 28 days. Her husband has fathered a child with another woman. Which cause of infertility is MOST likely?

A Polycystic ovary syndrome
B Hyperprolactinaemia
C Premature ovarian insufficiency
D Tubal damage from post-abortal pelvic infection
E Male factor, because it causes half of infertility

2. A 30-year-old woman with regular 35-day cycles is being investigated for infertility. On which day of her cycle should serum progesterone be measured to confirm ovulation?

A Day 2
B Day 28
C Day 21
D Day 14
E Day 35

3. A 25-year-old nulliparous woman with 2 years of primary infertility has cycles every 60–90 days, acne and hirsutism, and a BMI of 33. Prolactin and TSH are normal. Her husband's semen analysis is normal and HSG shows bilateral free spill. What is the MOST appropriate first-line pharmacological treatment, alongside lifestyle advice?

A Clomiphene 200 mg daily for 10 days
B Metformin alone for 2 years
C IVF immediately
D Laparoscopic ovarian drilling
E Letrozole 2.5 mg daily on days 3–7, increasing up to 7.5 mg if she does not ovulate

4. A 29-year-old woman attends a regional referral hospital asking for 'fertility tablets'. She has irregular cycles and 18 months of trying. A private clinic has already given her clomiphene for 8 months without follicle monitoring. Her husband has not had a semen analysis and she has had no tubal test. What is the MOST appropriate next step?

A Stop clomiphene; arrange semen analysis and HSG, and confirm ovulation, before planning further treatment
B Increase clomiphene to 250 mg daily
C Continue clomiphene for another 12 months
D Refer directly for IVF
E Start gonadotrophin injections

5. A husband's semen analysis (after 3 days of abstinence, delivered within 45 minutes) shows: volume 2.5 mL, concentration 22 million/mL, total motility 50%, progressive motility 35%, normal forms 5%, vitality 70%. How should this be interpreted?

A Oligozoospermia; refer for ICSI
B Asthenozoospermia; give testosterone
C All parameters are at or above the WHO 2021 lower reference limits
D Teratozoospermia; repeat in 1 week
E Azoospermia; karyotype

6. A 32-year-old woman's HSG shows bilateral distal tubal blockage with large, thin-walled hydrosalpinges. She and her husband (normal semen) can afford one cycle of IVF at a private centre in Kampala. What should be done BEFORE the IVF cycle to maximise success? Advanced

A Clomiphene for 6 months
B Open tuboplasty by laparotomy
C Hysteroscopic polypectomy
D Laparoscopic bilateral salpingectomy (or proximal tubal occlusion)
E Antibiotics for 6 weeks

7. A 34-year-old man has azoospermia on two semen analyses with normal volume. His testes are 20 mL each, both vasa are palpable, and FSH, LH and testosterone are normal. He had recurrent untreated urethral discharge in his twenties. What is the MOST likely cause? Advanced

A Klinefelter syndrome
B Obstructive azoospermia from post-infective epididymal blockage
C Hypogonadotropic hypogonadism
D Retrograde ejaculation
E Congenital bilateral absence of the vas deferens

8. A 28-year-old man who uses anabolic steroids for bodybuilding has a sperm concentration of 1 million/mL. His wife is 26 with regular cycles and patent tubes. What is the MOST appropriate first step?

A Proceed immediately to ICSI
B Prescribe testosterone replacement to improve sperm production
C Recommend donor sperm
D Varicocele surgery
E Stop the anabolic steroids and repeat the semen analysis after about 3 months

9. A 30-year-old woman with PCOS takes clomiphene 150 mg on days 2–6. Ultrasound tracking and mid-luteal progesterone show no ovulation over three cycles. Semen analysis and HSG are normal. Which is the MOST appropriate next option? Advanced

A Switch to letrozole (or add metformin), with gonadotrophins or laparoscopic ovarian drilling as further options
B Increase clomiphene to 300 mg
C Continue clomiphene 150 mg for 12 more months
D Hysterectomy
E Diagnose unexplained infertility and advise expectant management

10. A 31-year-old P1 has had scanty periods since a dilatation and curettage for retained placenta 2 years ago, complicated by fever. She has not conceived since. Hormone levels are normal. HSG shows irregular filling defects in the uterine cavity. What is the MOST likely diagnosis and treatment?

A PCOS; letrozole

B Premature ovarian insufficiency; egg donation

C Asherman's syndrome; hysteroscopic adhesiolysis followed by oestrogen and measures to prevent re-adhesion

D Submucous fibroid; myomectomy by laparotomy

E Cervical stenosis; cervical dilatation

11. A 29-year-old woman has secondary amenorrhoea, galactorrhoea and infertility. She takes no medications and is not pregnant. Serum prolactin is markedly raised on repeat testing; TSH is normal; MRI shows a 6 mm pituitary microadenoma. What is the BEST treatment to restore fertility?

A Clomiphene 50 mg days 2–6

B Trans-sphenoidal surgery

C Levothyroxine

D Cabergoline (or bromocriptine), titrated to normalise prolactin

E Combined oral contraceptive pill

12. Five days after hCG trigger in a gonadotrophin cycle, a 27-year-old woman with PCOS presents with abdominal distension, nausea and breathlessness. Ultrasound shows enlarged ovaries (10 cm) and moderate ascites. Haematocrit is 48% and urine output is low. Which is the MOST appropriate management? Advanced

A Discharge with ibuprofen for pain

B Admit: strict fluid balance, IV fluids to maintain urine output, LMWH thromboprophylaxis, and paracentesis if ascites becomes tense or breathing is compromised

C Immediate laparotomy and oophorectomy

D Diuretics (furosemide) to remove the ascites

E Bimanual examination to assess ovarian size

13. An HIV-serodiscordant couple wants to conceive. The man is HIV-positive, has taken TLD for 4 years and his viral load has been undetectable for 2 years; the woman is HIV-negative. Both are otherwise fertile. What is the MOST appropriate advice? Advanced

A They must never have condomless intercourse; IVF is the only option

B The woman should start ART herself

C Adoption is the only safe option

D Sperm washing and IUI are mandatory

E With a sustained undetectable viral load he does not transmit HIV sexually; offer PrEP to her if desired, time condomless intercourse to the fertile window, and continue viral load monitoring

14. An infertile woman's HSG is planned. Her LMP started 3 days ago and bleeding is ongoing. She had PID 2 years ago. Which plan is MOST appropriate?

A Schedule the HSG for day 6–10 of this cycle (after bleeding stops, before ovulation), with chlamydia screening or antibiotic prophylaxis

B Do the HSG today

C Do the HSG on day 20 of the cycle

D HSG is contraindicated because of previous PID

E Replace HSG with a day-21 progesterone

15. A couple with 3 years of infertility have normal ovulation, patent tubes on HSG, a normal uterine cavity and normal semen analysis. The woman is 31. They ask what they can do next. Which response is BEST?

A Clomiphene alone is the recommended treatment for unexplained infertility

B Stop trying; unexplained infertility means they are sterile

C Explain unexplained infertility: many couples still conceive naturally, but after 2 or more years of trying IVF is the most effective treatment; stimulated IUI may be a lower-cost alternative where IVF is unaffordable

D Laparotomy to look for hidden causes

E Test the man's testosterone and give testosterone if low-normal

Answers & explanations

1. D · Tubal damage from post-abortal pelvic infection

D is correct. Secondary infertility after **post-abortal sepsis**, with regular (ovulatory) cycles, points to **tubal damage** (and possibly Asherman's) — the commonest pattern in Uganda. **A**, **B** and **C** are wrong: all cause irregular cycles or amenorrhoea, and she cycles regularly. **E** is wrong: male factor must still be tested (previous paternity does not exclude a new problem), but the history makes tubal disease the most likely cause.

2. B · Day 28

B is correct. Progesterone peaks about **7 days after ovulation**, i.e. **7 days before the next period**. In a 35-day cycle that is **day 28**. **C** (day 21) is only correct for a 28-day cycle — a common error that gives falsely 'low' results. **A** is used for FSH/LH/oestradiol. **D** is around ovulation in a 28-day cycle, before the luteal peak. **E** is the day the next period is expected, when progesterone has fallen.

3. E · Letrozole 2.5 mg daily on days 3–7, increasing up to 7.5 mg if she does not ovulate

E is correct. She has **PCOS** (oligo-ovulation + hyperandrogenism), with patent tubes and normal semen. **Letrozole** is first-line (higher ovulation and live-birth rates than clomiphene), together with weight loss and folic acid. **A** is wrong: clomiphene starts at 50 mg for 5 days and the maximum is 150 mg. **B** is wrong: metformin alone is less effective and 2 years wastes time. **C** is wrong: IVF is not first-line for anovulatory PCOS. **D** is a second-line option for clomiphene-resistant PCOS.

4. A · Stop clomiphene; arrange semen analysis and HSG, and confirm ovulation, before planning further treatment

A is correct. Ovulation induction must **not** be given without knowing the **semen** is adequate and the **tubes** are patent; clomiphene should not exceed **6 months** and needs **ultrasound monitoring**. **B** exceeds the maximum dose (150 mg). **C** prolongs inappropriate unmonitored treatment. **D** may eventually be needed but is premature without a diagnosis. **E** carries high OHSS/multiple-pregnancy risk and still skips basic assessment.

5. C · All parameters are at or above the WHO 2021 lower reference limits

C is correct. WHO 2021 limits: volume 1.4 mL, concentration 16 million/mL, total motility 42%, progressive 30%, normal forms 4%, vitality 54% — every value here is above them. **A** is wrong: 22 million/mL is not low. **B** is wrong on two counts: motility is normal, and testosterone would suppress spermatogenesis. **D** is wrong: 5% normal forms exceeds 4%. **E** is wrong: sperm are present.

6. D · Laparoscopic bilateral salpingectomy (or proximal tubal occlusion)

D is correct. Hydrosalpinx fluid is **embryotoxic** and washes embryos out; removing or clipping the tubes roughly **doubles IVF success**. **A** is wrong: her problem is tubal, not ovulatory. **B** is wrong: tuboplasty for large hydrosalpinges has poor results, and she has already chosen IVF. **C** is wrong: there is no polyp. **E** is wrong: antibiotics do not reopen damaged tubes or remove the fluid.

7. B · Obstructive azoospermia from post-infective epididymal blockage

B is correct. Normal testicular volume and normal FSH mean sperm production is intact, so azoospermia is due to **obstruction** — here most likely epididymal scarring after gonococcal/chlamydial epididymitis. **A** is wrong: Klinefelter causes small firm testes and high FSH. **C** is wrong: gonadotrophins and testosterone would be low. **D** is wrong: retrograde ejaculation gives a low-volume or absent ejaculate. **E** is wrong: both vasa are palpable (and CBAVD usually gives low volume, acidic semen).

8. E · Stop the anabolic steroids and repeat the semen analysis after about 3 months

E is correct. Anabolic steroids suppress FSH/LH and spermatogenesis; stopping them usually allows recovery over months (spermatogenesis takes about 74 days, so repeat after about 3 months or later). **A** and **C** are premature for a probably reversible cause. **B** is wrong: testosterone would worsen the suppression. **D** is wrong: there is no varicocele, and NICE does not recommend varicocele repair as a fertility treatment.

9. A · Switch to letrozole (or add metformin), with gonadotrophins or laparoscopic ovarian drilling as further options

A is correct. She is **clomiphene-resistant** (no ovulation at 150 mg). Options: **letrozole**, clomiphene + **metformin**, low-dose **gonadotrophins** with monitoring, or **laparoscopic ovarian drilling**. **B** exceeds the maximum dose. **C** continues an ineffective treatment beyond the 6-month limit. **D** is absurd. **E** is wrong: the cause (anovulation) is known.

10. C · Asherman's syndrome; hysteroscopic adhesiolysis followed by oestrogen and measures to prevent re-adhesion

C is correct. Hypomenorrhoea after curettage of a recently pregnant, infected uterus with cavity filling defects = **Asherman's syndrome** (intrauterine adhesions). Treatment: **hysteroscopic adhesiolysis**, then oestrogen and a temporary balloon/IUD to prevent re-adhesion. **A** and **B** are wrong: hormones are normal. **D** is wrong: a fibroid causes a smooth filling defect and usually heavier, not scantier, periods. **E** is wrong: stenosis causes cyclical pain/haematometra and HSG cannula difficulty, not intracavitary defects.

11. D · Cabergoline (or bromocriptine), titrated to normalise prolactin

D is correct. Prolactin suppresses GnRH pulsatility; a **dopamine agonist** (cabergoline 0.25 mg twice weekly, or bromocriptine 1.25 mg at night titrated up) normalises prolactin and restores ovulation in most women and shrinks the adenoma. **A** is wrong: clomiphene will not overcome prolactin suppression. **B** is reserved for drug failure/intolerance. **C** is wrong: TSH is normal. **E** is wrong: it prevents pregnancy.

12. B · Admit: strict fluid balance, IV fluids to maintain urine output, LMWH thromboprophylaxis, and paracentesis if ascites becomes tense or breathing is compromised

B is correct. This is **severe OHSS** (haemoconcentration, oliguria, ascites). Management is supportive: admission, fluid balance, IV fluids, **thromboprophylaxis** (high thrombosis risk), analgesia without NSAIDs, and **paracentesis** for tense ascites. **A** is wrong: NSAIDs impair renal function and she needs admission. **C** is wrong: surgery is only for torsion/rupture/haemorrhage. **D** is wrong: diuretics worsen intravascular depletion and thrombosis. **E** is wrong: vigorous examination may rupture the enlarged, fragile ovaries.

13. E · With a sustained undetectable viral load he does not transmit HIV sexually; offer PrEP to her if desired, time condomless intercourse to the fertile window, and continue viral load monitoring

E is correct. U = U: a partner with a sustained undetectable viral load on ART does not transmit HIV sexually. Safer conception: confirm suppression, offer **PrEP** to the negative partner if she wishes, limit condomless sex to the **fertile window**, continue monitoring and folic acid. **A**, **C** and **D** are unnecessarily restrictive and costly. **B** is wrong: an HIV-negative person takes PrEP, not ART.

14. A · Schedule the HSG for day 6–10 of this cycle (after bleeding stops, before ovulation), with chlamydia screening or antibiotic prophylaxis

A is correct. HSG is done after menses and **before ovulation** (days 6–10) to get a clear image, avoid pushing menstrual debris/infection, and avoid irradiating a possible early pregnancy. With a past history of PID, screen for chlamydia or give **prophylactic antibiotics**. **B** is wrong: during menstruation the images are poor and the infection risk higher. **C** is wrong: after ovulation she could be pregnant. **D** is wrong: only **active** pelvic infection is a contraindication (laparoscopy and dye is an alternative if PID history is significant). **E** is wrong: progesterone assesses ovulation, not the tubes.

15. C · Explain unexplained infertility: many couples still conceive naturally, but after 2 or more years of trying IVF is the most effective treatment; stimulated IUI may be a lower-cost alternative where IVF is unaffordable

C is correct. In **unexplained infertility**, NICE advises IVF after a total of 2 years of trying; oral ovulation induction alone is not recommended. In low-resource settings stimulated IUI is often used as a cheaper intermediate step. **A** is wrong: clomiphene alone does not improve live births in unexplained infertility. **B** is wrong and harmful: they are not sterile. **D** is wrong: laparotomy is not a diagnostic tool here (laparoscopy may be considered in selected women). **E** is wrong: testosterone suppresses spermatogenesis.

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: Secondary infertility after a septic abortion

Mr and Mrs Okello attend the gynaecology clinic at a regional referral hospital. Mrs Okello, 28, P1+1, has not conceived for 3 years. Her second pregnancy ended at 10 weeks in an incomplete abortion evacuated at a private clinic, followed by fever and pain. Her mother-in-law is pressuring her husband to take a second wife.

1. What focused history do you take from each partner, and why?

> **Her**: menstrual pattern (regular → likely ovulatory; scanty → **Asherman's**), post-abortal fever/pain (tubal damage), STIs/PID and partner treatment, previous surgery, dysmenorrhoea/dyspareunia, weight/hirsutism, galactorrhoea, contraception (DMPA), HIV status, coital frequency, psychosocial pressure and **violence**. **Him**: previous children, mumps orchitis, undescended testes, urethral discharge/epididymitis, sexual function, alcohol/smoking/steroids/heat, chronic illness, HIV.

2. Her cycles are regular (28 days) with normal flow. Pelvic exam: fixed retroverted uterus, bilateral adnexal thickening, mild tenderness. His testes are 18 mL, vasa present, no varicocele. What are the likely diagnosis and differentials?

> Secondary infertility, most likely **tubal-peritoneal factor** from **post-abortal PID** (regular cycles suggest ovulation). Differentials: Asherman's syndrome (less likely with normal flow), **male factor** (must be tested), endometriosis, genital TB, coexisting chronic PID.

3. Outline the investigations in the order you would do them.

> 1. **Semen analysis** (cheap, non-invasive) — WHO 2021 limits. 2. **Mid-luteal progesterone** on day 21 (>30 nmol/L). 3. HIV, syphilis, HBV for both; chlamydia/gonorrhoea test if available; Hb. 4. **Pelvic ultrasound** (hydrosalpinx, fibroids). 5. **HSG** on day 6–10 with **antibiotic prophylaxis** (e.g. doxycycline) because of past PID — or laparoscopy and dye where available (allows adhesiolysis at the same time).

4. Semen is normal, progesterone 42 nmol/L, and HSG shows bilateral hydrosalpinges with no spill. What are the options, and how do you counsel them?

> Severe distal tubal disease: natural conception is unlikely and tubal surgery has poor results. The effective option is **IVF**, after **laparoscopic salpingectomy/proximal occlusion** of the hydrosalpinges (hydrosalpinx fluid halves IVF success). Discuss cost, success rates (age-dependent), private Kampala centres and the public IVF service at Mulago Specialised Women and Neonatal Hospital; **adoption/fostering**; and **ectopic risk** in any pregnancy. Counsel **together**: explain that the cause is past infection, not a curse or her 'fault'; address the family pressure, screen for depression and violence, and involve a counsellor.

5. How would you summarise this couple on a ward round?

> 'Mr and Mrs O: 3 years of secondary infertility after a post-abortal infection. She is ovulating (progesterone 42), his semen is normal, and HSG shows bilateral hydrosalpinges. Diagnosis: tubal factor infertility. Plan: counsel about IVF with prior laparoscopic salpingectomy, discuss costs and alternatives including adoption, psychosocial support, folic acid, and early ultrasound in any future pregnancy because of ectopic risk.'

Take-away: In Uganda, secondary infertility after abortion or delivery is tubal until proven otherwise: test the semen first, confirm ovulation, image the tubes safely, and counsel the couple honestly and together — prevention of PID is the real cure.

Case 2: Irregular periods and hirsutism

Ms Nansubuga, a 24-year-old married teacher from Masaka, presents with 2 years of primary infertility. Her periods come every 2–4 months, she has acne and facial hair she shaves weekly, and she has gained 15 kg since marriage. BMI 34.

1. What is your provisional diagnosis and what must you exclude?

> **PCOS** (oligo-anovulation + clinical hyperandrogenism). Exclude **hyperprolactinaemia**, **thyroid disease**, **pregnancy**, non-classic **CAH** (17-OH progesterone if very hirsute/early onset) and **androgen-secreting tumour** (rapid virilisation, very high testosterone).

2. What investigations do you order?

> Pregnancy test; **TSH, prolactin**; total testosterone (± SHBG); day 2–5 **FSH/LH** (LH often raised); **pelvic ultrasound** (≥20 follicles per ovary or volume ≥10 mL; endometrial thickness); fasting glucose/HbA1c and lipids; **husband's semen analysis**; **HSG** (tubal patency) before ovulation induction; progesterone is unhelpful while she has no predictable cycle.

3. Results: TSH and prolactin normal, testosterone mildly raised, polycystic ovaries on ultrasound, normal semen, bilateral spill on HSG. Write the management plan.

> 1. **Lifestyle**: diet and exercise targeting **5–10% weight loss** (can restore ovulation). 2. **Folic acid** (5 mg daily as BMI ≥30). 3. Induce a withdrawal bleed if needed (e.g. medroxyprogesterone 10 mg daily for 10 days). 4. **Letrozole 2.5 mg PO daily days 3–7**; confirm ovulation with **mid-luteal progesterone** (7 days before expected period) ± follicle tracking; increase by 2.5 mg per cycle up to **7.5 mg** if no ovulation. 5. Alternative if letrozole unavailable: **clomiphene 50 mg days 2–6** (max 150 mg, monitored, ≤6 months) ± **metformin**. 6. Screen for diabetes; manage hirsutism later (COC is not compatible with trying to conceive).

4. She does not ovulate on letrozole 7.5 mg or clomiphene + metformin. What are the next options and their risks?

> **Low-dose step-up gonadotrophins** with ultrasound monitoring (risk: **OHSS, multiple pregnancy** — cancel if >2–3 mature follicles), or **laparoscopic ovarian drilling** (restores ovulation without multiple-pregnancy risk; risk of adhesions and reduced ovarian reserve), then **IVF** if these fail (WHO 2025 suggests IVF rather than expectant management after failed pharmacological treatment).

Take-away: In PCOS, exclude mimics, confirm the tubes and semen are normal, start with weight loss and letrozole, confirm ovulation every cycle, and escalate stepwise while guarding against multiple pregnancy and OHSS.

Case 3: The man who would not give a sample

Mrs Atim, 30, has been trying to conceive for 4 years. She has had two HSGs (both normal), regular cycles, a mid-luteal progesterone of 38 nmol/L, and 6 months of clomiphene from a private clinic. Her husband, 36, a long-distance truck driver, has refused semen analysis, saying 'I am a man, the problem is hers.'

1. What is the most important missing piece of the evaluation and how will you approach it?

> The **semen analysis**. Invite him to the clinic, see the couple together, explain respectfully that about **half of infertility involves the man**, that the test is simple and confidential, and that it may save his wife unnecessary drugs and procedures. Explain collection (2–7 days' abstinence, masturbation into a sterile container, to lab within 1 hour).

2. He agrees. History: sits for long hours in a hot cab, drinks heavily at weekends, smokes, had 'swollen painful testicles' after a urethral discharge 10 years ago. Semen: volume 2.0 mL, concentration 3 million/mL, progressive motility 15%, normal forms 2%. Interpret.

> Severe **oligo-astheno-teratozoospermia** (all below WHO 2021 limits: 16 million/mL, 30% progressive, 4% normal forms). Possible causes: previous **epididymo-orchitis** (partial obstruction/testicular damage), **heat**, alcohol, smoking. Repeat the analysis in about 3 months.

3. What further assessment and initial management do you arrange?

> Examine: testicular volume, vasa, epididymal thickening, **varicocele**. Tests: **FSH, LH, testosterone**; scrotal ultrasound; if concentration <5 million/mL consider **karyotype and Y-microdeletion** testing. Treat any STI in both. Lifestyle: stop smoking, reduce alcohol, loose clothing and breaks from heat, weight; **no testosterone or 'strength' supplements**; repeat semen analysis after 3 months. Stop the wife's clomiphene (she ovulates).

4. Repeat analysis 3 months later is unchanged; FSH is mildly raised. What treatment options do you discuss?

> With severe male factor, the most effective option is **IVF with ICSI** (private Kampala centres or Mulago public IVF, cost permitting). Alternatives: **donor sperm** (cultural/religious acceptability varies) or **adoption/fostering**. IUI is unlikely to succeed with such low motile counts. Provide couple counselling, protect the husband's dignity and confidentiality, and address the burden previously placed on his wife.

Take-away: Semen analysis is the cheapest and most neglected test in infertility — do it early, engage the man with respect, and never give the woman ovulation induction when she already ovulates.

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.